



Class Review

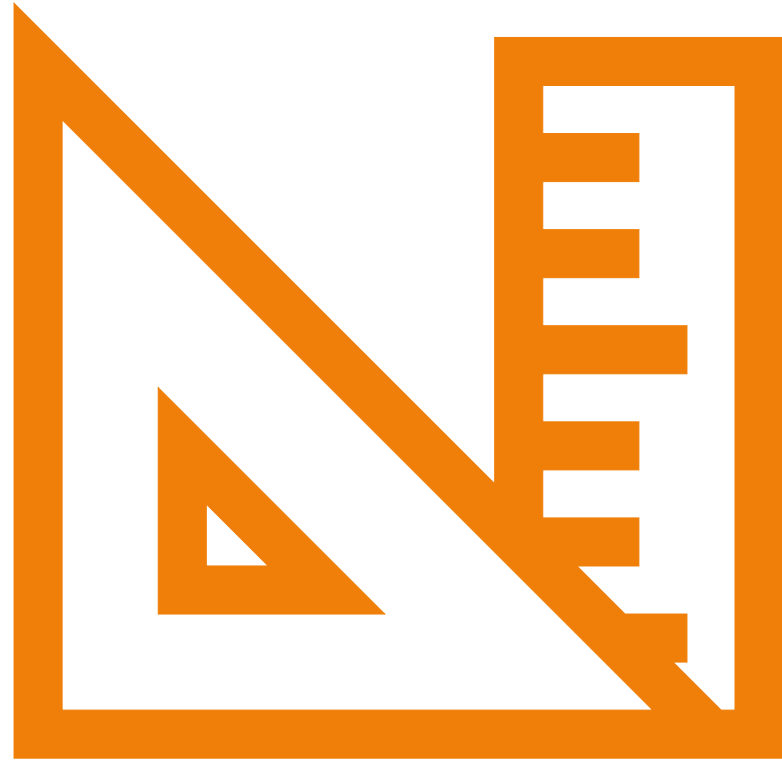
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Class Roadmap

- What are Research Methods
- Getting Ready to Do Statistical Analysis
- Describing Data
- Univariate Distributions
- Bivariate Distributions
- Probability
- Binomials and other Sampling Distributions
- Hypothesis Testing
 - T-Tests
 - F-Tests
- Causal Inference
- Experimental Designs
- Ethics
- Reproducibility
- Statistical Thinking



Research Design

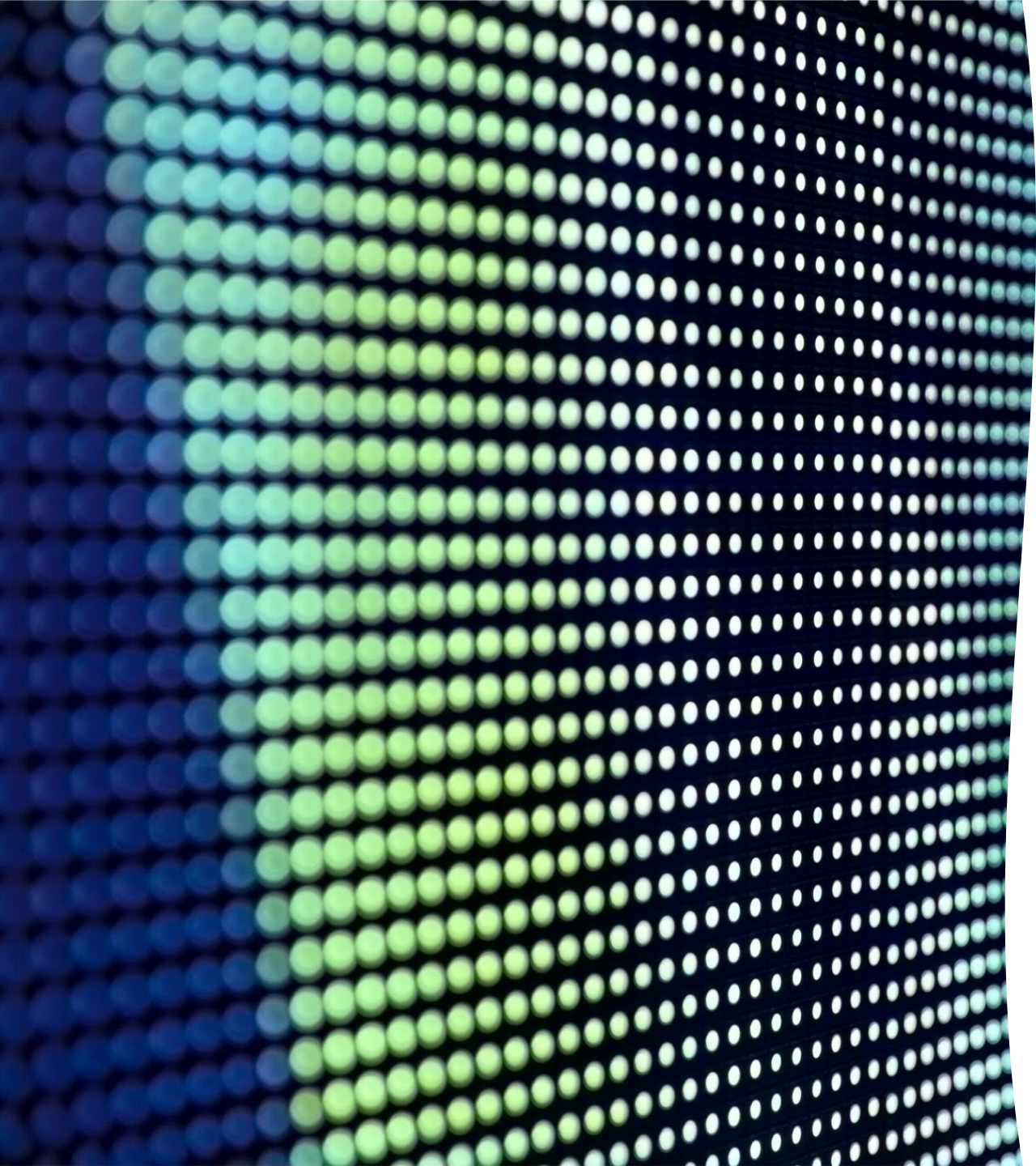


Every baby
knows the
scientific method!



Steps in a Research Study

- Question
- Design
- Sampling
- Measurement
- Summary



Asking a Good Question

Research Problem

Research Question

Operationalize

Hypothesize

Good questions begin with problems

- Research problems are broad issues or situations that we'd like to
 - describe,
 - understand, or
 - change.



Research problems

- Broadly, these include:
 - Areas of concern,
 - Conditions that could be improved,
 - Difficulties that need to be eliminated, and
 - Questions seeking answers.

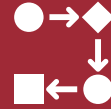


Research Questions

- A research problem leads to a research question
- Questions should in some way. . .
 - Be worth investigating,
 - Contribute knowledge to the field,
 - Improve educational practice, or
 - Improve the human condition.



Operationalize



Operationalization is a process of defining the fuzzy concepts; and



making the concept(s)

distinguishable,
measurable, and
understandable in terms of
observable data.



Determine which variables you would like to understand.



You will want to be specific.

Hypothesizing



A hypothesis is a

precise, testable statement of
what the researchers predict will be the
outcome of the study.



In other words:

What do you think the answer is,

- given what we already know?



Research Design

- 2nd step in a research study
- Three types
 - Randomized experiment (true experiment)
 - Quasi-experiment (non-randomized experiments)
 - Observational Design (non-experimental)

Randomized Experiment (strong research design)

- The only design that strongly supports causal inference

Two things are related (correlated)

- A caused B
- B caused A
- C caused both A and B
 - Names for C
 - Lurking Variable,
 - Confounding Variable,
 - Third Variable
 - Spurious Relationship

Background and History

- Random assignment
 - equates groups on all confounding variables ON THE AVERAGE (across replications)
 - Treatment Group
 - Control Group



Variables



Definition: A feature of an individual that can take on different values



Conceptual Variable

Def: Researcher's specification of the topic of study



Operational Definition

Def: Links the conceptual variable to one that can be measured.





Variables



Explanatory Variable

Def: variable measured or manipulated by the researcher; the possible cause

Same as independent variable



Response Variable/ Dependent Variable

Def: outcome of interest



Lurking / Confounding Variable

Def: Uncontrolled variable that hides the relationship between the explanatory and the response variable.



Quasi- Experimental Design

- Non-random experiment
- Def. Has all the features of a randomized experiment, except random assignment
- Reasons why we can't do a randomized experiment?



Quasi-Experimental Design

- Reasons why we can't do a randomized experiment?
 - May be logically impossible because subjects show up with their explanatory variable features defined
 - Ethically problematic
 - Can't withhold treatment
 - Can't inflict harm
 - Impractical
 - Too long for data collection
 - Too expensive to collect data
 - 1 time event
- We worry more about threats to validity

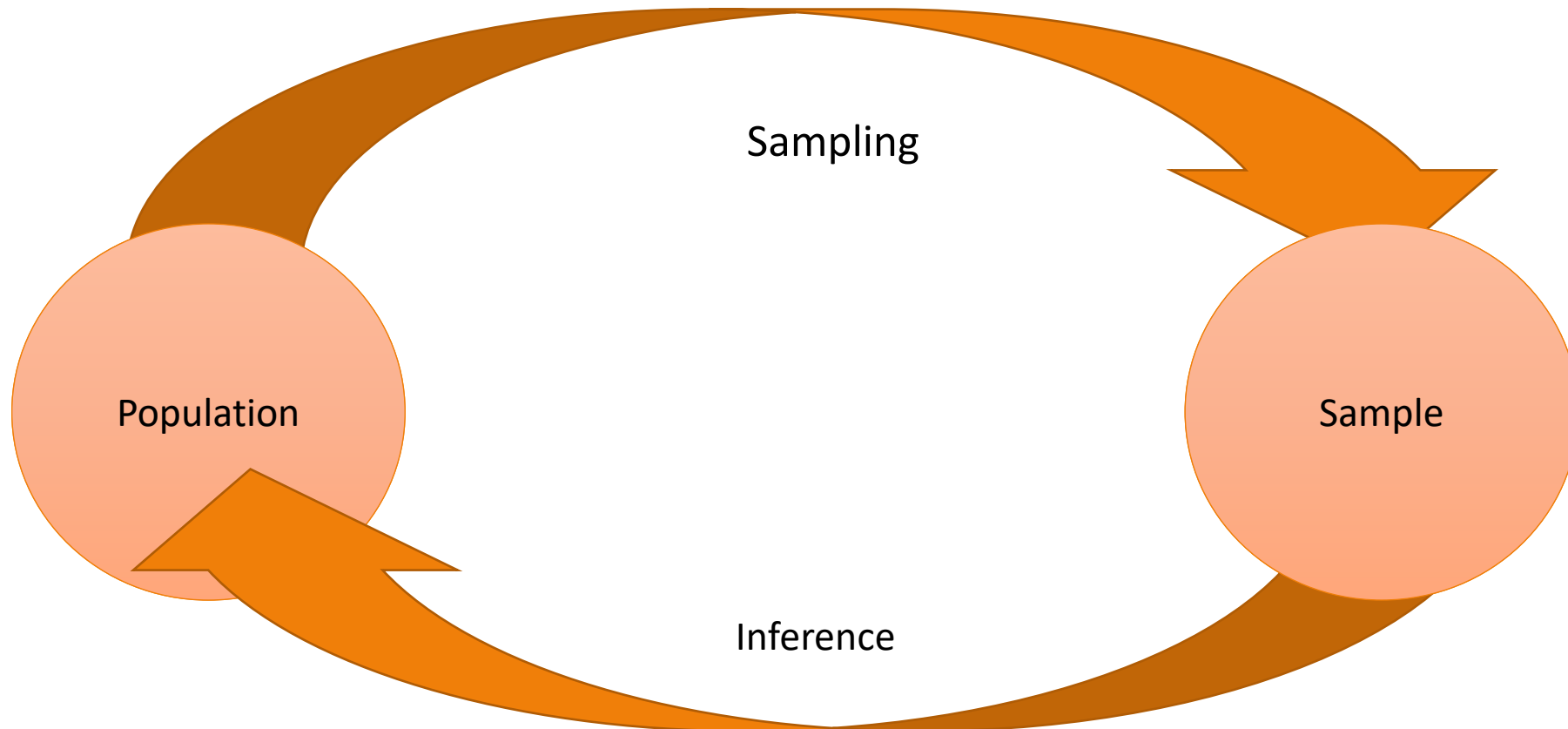


Sampling Theory



Sampling

- Third Step in Study
 - Collect units to study



Definitions

Population

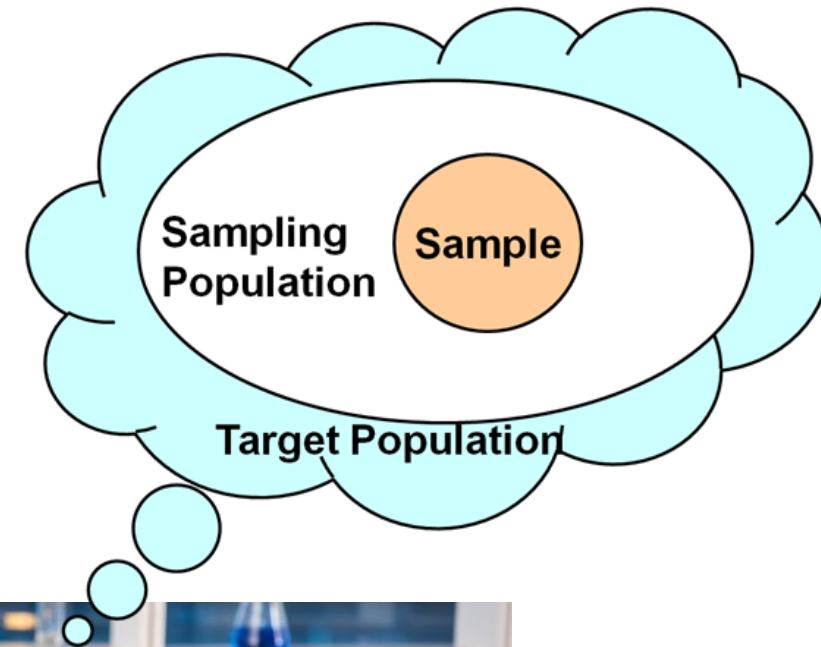
- entire group about whom we want information

Sample

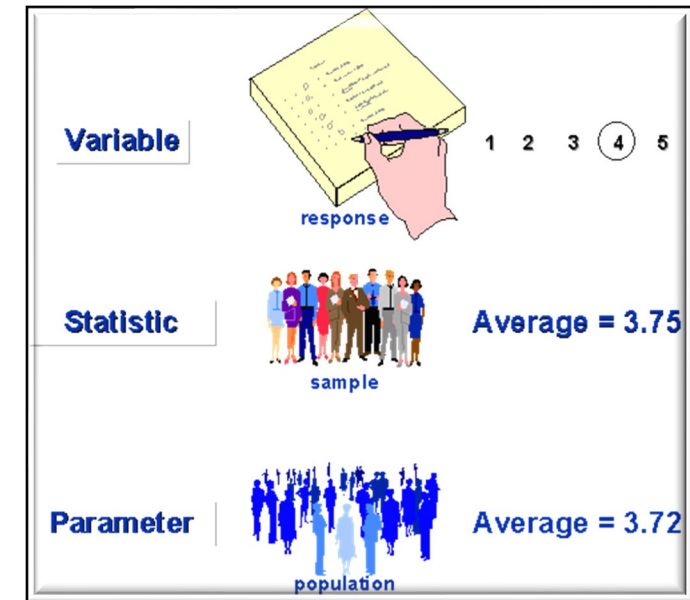
- part of a population from which we collect information

Sampling frame

- the list of units from which samples are chosen

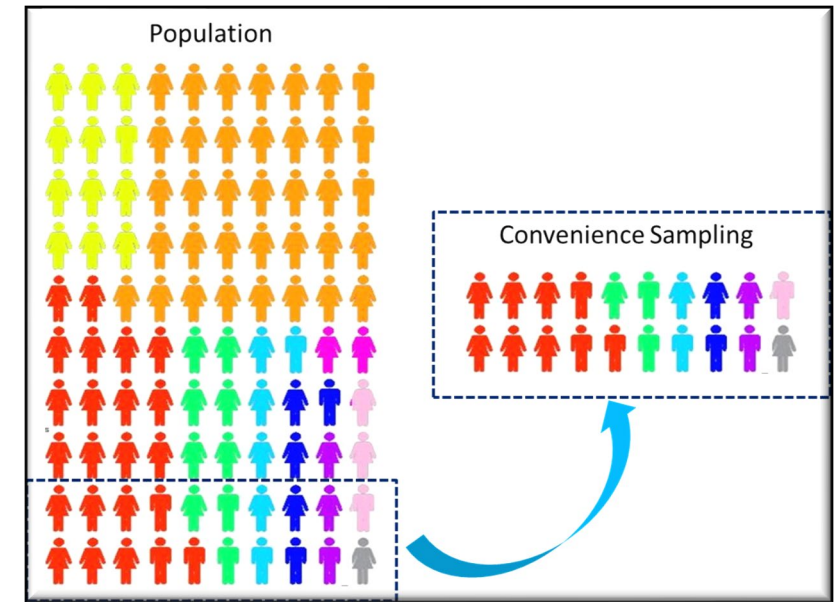


Definitions



- Parameter
 - number that is associated with a population
- Statistic
 - a number that describes a sample
 - We use statistics to estimate parameters
- We draw inferences from a sample to the population

Convenience sample



- Convenience sample
 - Use an available but non-representative set of respondents
 - Examples
 - In the Classroom
 - Research Driven Sampling
 - Growing Trends
 - Biological Surveys



Representative sample

- We want a sample to represent a population
 - If not, how does the sample differ from the population?
- Sampling bias leads to systematic errors
 - how a sample represents the population

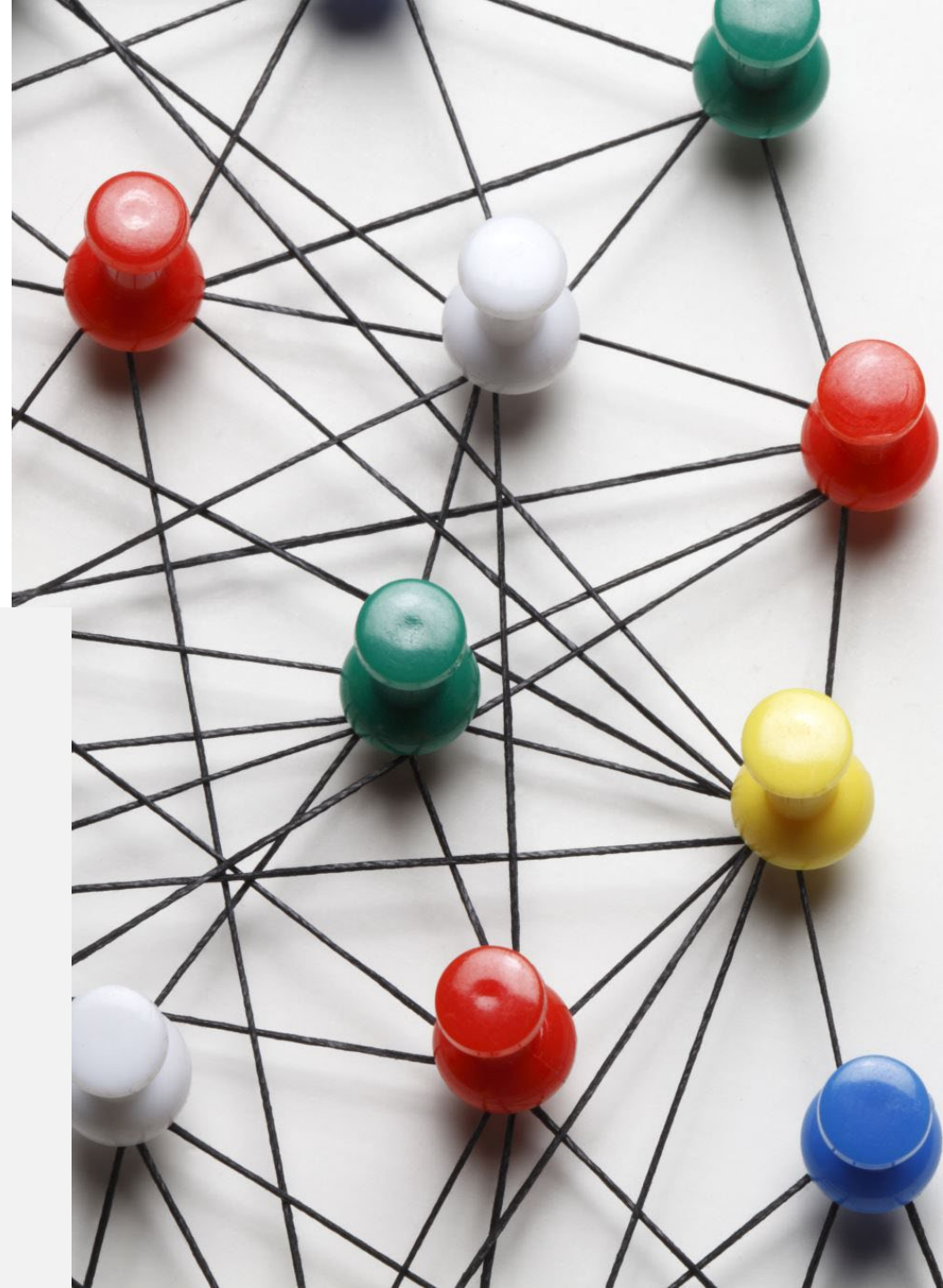


Table of the Elements

Number → 1

Symbol ← H

Name → Hydrogen

Atomic Weight ← 1.008

																		13 IIIA 5 B Boron 10.81	14 IVA 6 C Carbon 12.01	15 VA 7 N Nitrogen 14.007
																		13 Al Aluminium 26.9815385	14 Si Silicon 28.085	15 P Phosphorus 30.973761998
23 5B V Vanadium 50.9415	24 6VB Cr Chromium 51.9961	25 7VIIB Mn Manganese 54.938044	26 8VIIIB Fe Iron 55.845	27 9VIIIB Co Cobalt 58.933194	28 10VIIIB Ni Nickel 58.6934	29 11IB Cu Copper 63.546	30 12IIB Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.921595										
41 Nb Niobium 92.90637	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.90550	46 Pd Palladium 106.42	47 Ag Silver 107.8682	48 Cd Cadmium 112.414	49 In Indium 114.818	50 Sn Tin 118.710	51 Sb Antimony 121.760										
73 Ta Tantalum 180.94788	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.084	79 Au Gold 196.966569	80 Hg Mercury 200.592	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98040										
105 Db Dubnium (261)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (269)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)										
120 Dl Darmstadtium (310)	121 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.92535	66 Dy Dysprosium 162.500	67 Ho Holmium 164.93032	68 Er Erbium 167.259	69 Tm Thulium 168.93032	70 Yb Ytterbium 173.054										
130 Nh Nihonium (286)	131 Ds Darmstadtium (281)	132 Rg Roentgenium (282)	133 Cn Copernicium (285)	134 Nh Nihonium (286)	135 Fl Flerovium (289)	136 Mc Moscovium (290)	137 Lv Livermorium (293)	138 Ts Tennessine (294)	139 Og Oganesson (294)	140 Lr Lawrencium (260)										
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190 Nh Nihonium (286)	191 Ds Darmstadtium (281)	192 Rg Roentgenium (282)	193 Cn Copernicium (285)	194 Nh Nihonium (286)	195 Fl Flerovium (289)	196 Mc Moscovium (290)	197 Lv Livermorium (293)	198												

Random Sampling

- Best way to ensure a representative sample
- Probability sample
 - a sample in which each element has a known (a priori) probability of being drawn
- Simple random sample (SRS)
 - every sample of a given size has an equal probability of being chosen.
- Random mechanisms
 - Random digit table (*Rand Corporation 1955*)
 - Computer generated random numbers (used to be call pseudorandom)



No one random sample is perfectly representative.

- 
- They are representative **on average**
 - We can expect the pattern across samples -> sampling distribution
 - The bigger the sample, the closer it is to the population, the more representative
 - When sample = population , census
 - Sampling fraction
 - n denotes sample size
 - N denotes population size
 - Sampling fraction = n/N

Other Sampling designs

- Multistage samples
 - Also called cluster samples
 - We don't have a sampling frame of units of interest
 - We sample higher level units that do have sampling frames
 - Example
 - Sample 5 of 50 states
 - Samples 100 census tracks in each state
 - Sample 10 people from each census track
 - $n=5,000$



Other Sampling designs

- Stratified random sampling
 - Goal: to increase precision of estimates that come from sample
 - Mechanism – use strata in which responses are more similar to one another than average
 - We can use multistage sampling AND strata random sampling in the same study
-



Election of 1948

Use of Quota sampling

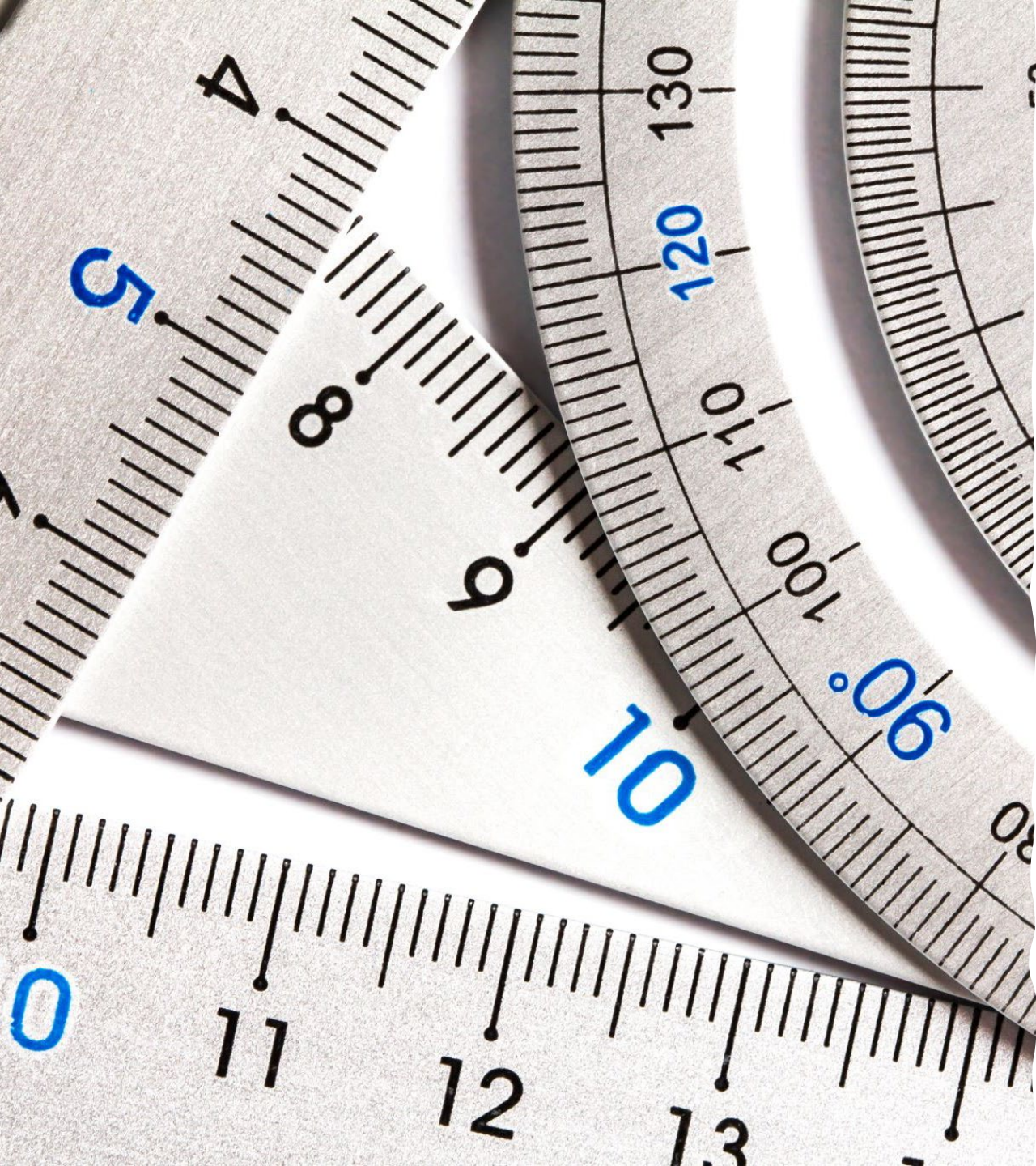
- introduced by George Gallup as early as 1935
- successfully used by him to predict the winner of the 1936, 1940 and 1944 elections.

Quota sampling is a systematic effort to

- force the sample to fit a certain national profile by using quotas

The sample should have

- so many women, so many men, etc
- represent the same proportions in the sample as are in the electorate at large.



Measurement

Measurement

4th step in the research process

- Def: assignment of numbers to characteristics of people or objects
- Specialties: psychometrics; econometricians, biometricians; sociometrists
- Examples: measurement scales
 - Speed – > miles-per-hour
 - Temperature (kelvin)
 - Order of finishers
 - Numbers on the back of basketball jerseys

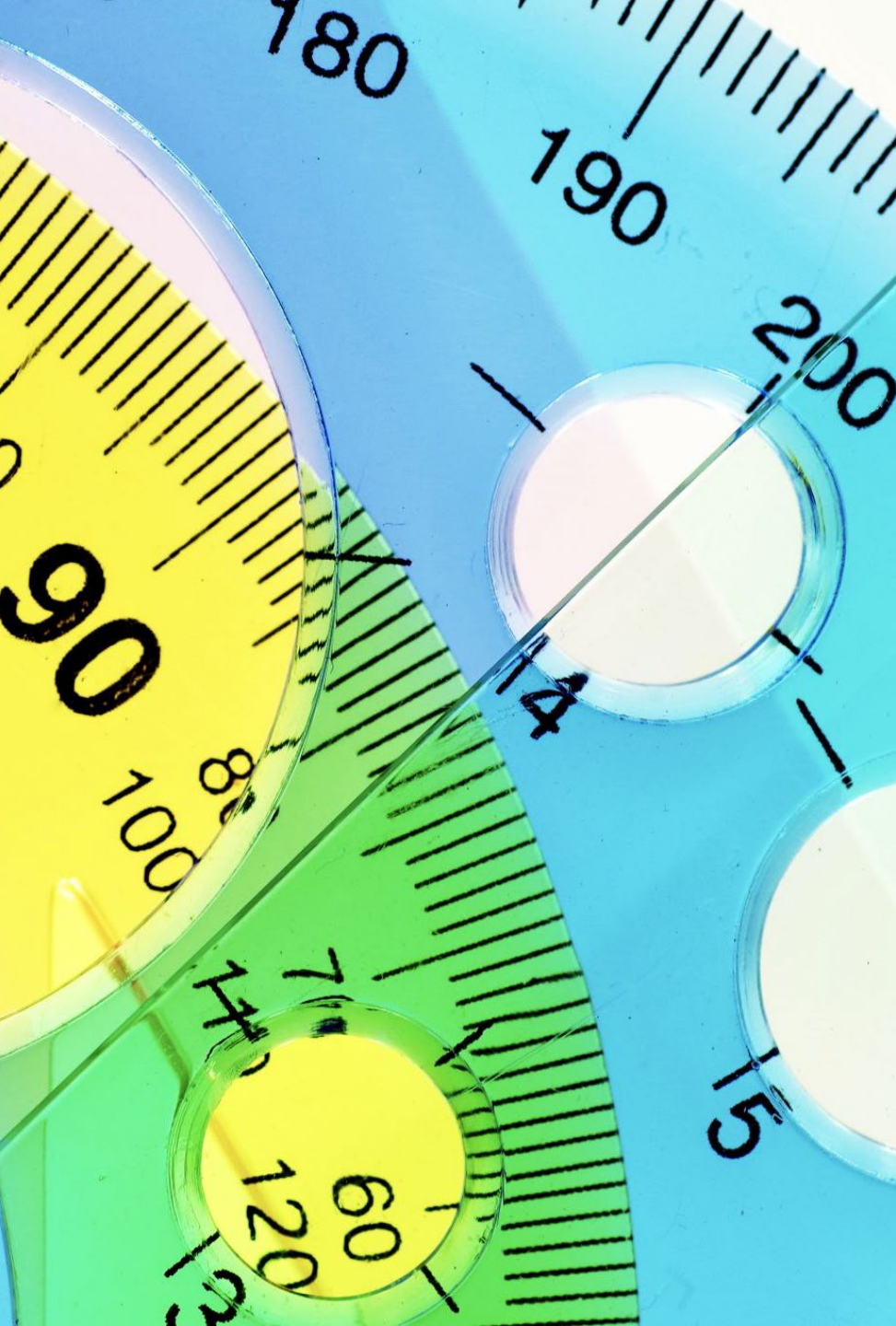
Variables contain the outcome of measurement processes

- Qualitative: number represent qualities (not quantities)
- Quantitative: numbers mean something in relation quantities in the real world



Measurement

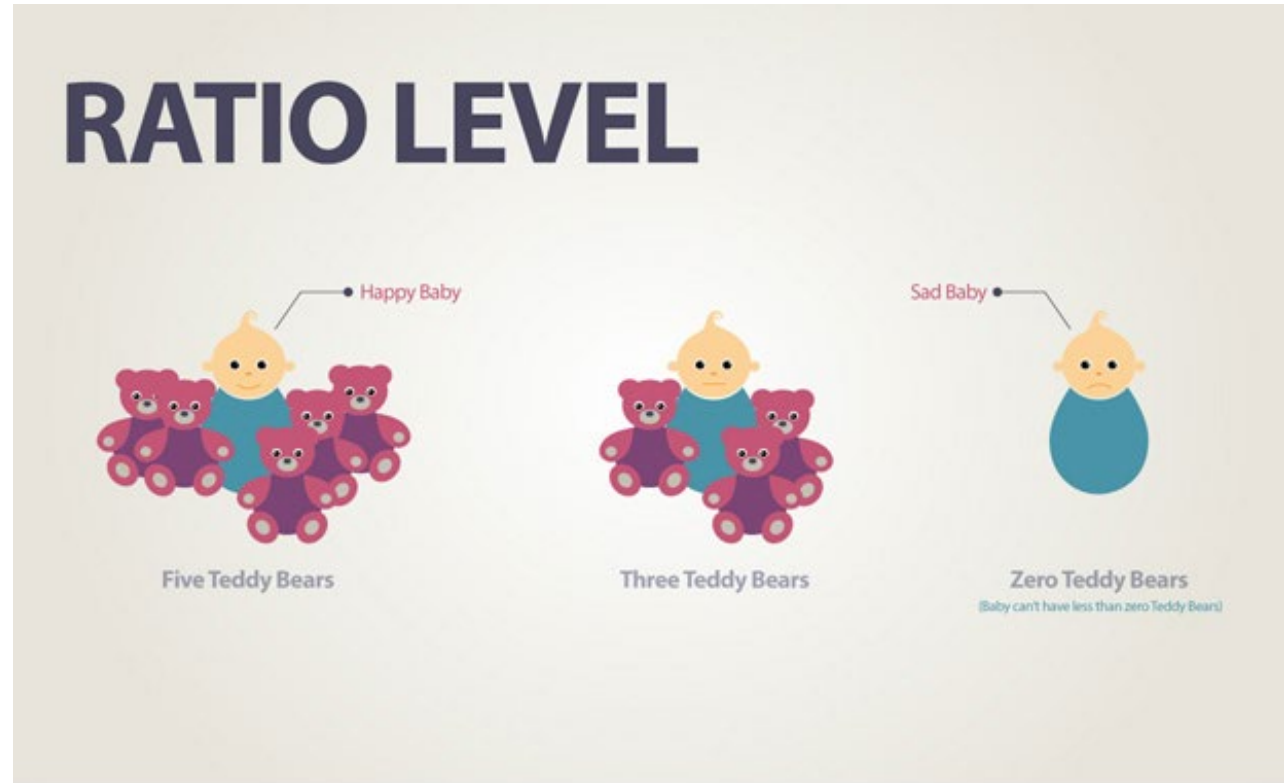
- Stevens (1946) Levels of Measurement
 - Nominal
 - Ordinal
 - Interval
 - Ratio



Ratio Features

- Have absolute zero
 - 0 indicated absence (origin means zero)
- Equal intervals
 - An interval means the same value at any point on measurement scales
- Order
 - Number means order
- Identity
 - Different numbers mean different measurement outcomes ($1 \neq 2$)

Ratio (quantitative)



- All four features
- Multiplication is a permissible transformation

Interval (quantitative)

- *Has three features (all but absolute zero)*
- Addition is a permissible transformation
- Temperature in Fahrenheit is an example

d) Estimate the heart rate when temperature is 98.7°C .

0 bpm

Ordinal (qualitative)

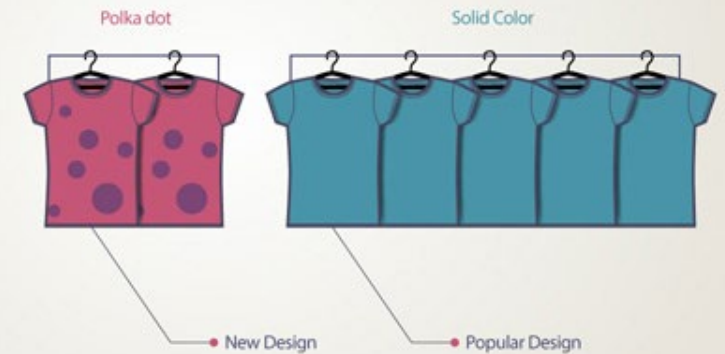
- Has order and identity
- Monotonic transformations are permissible
 - These maintain the order of the values



Nominal (qualitative)

- Only has identity
- Any identity preserving transformation is permissible

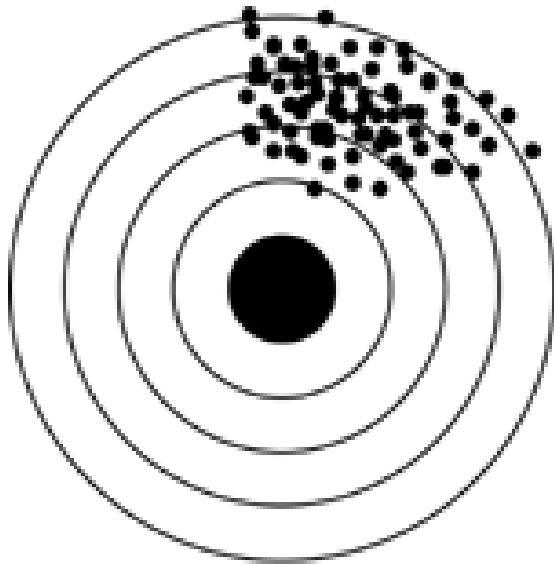
NOMINAL DATA



Goals of Measurement

- Reliable
 - *“...the degree to which a test or measure produces the same scores when applied in the same circumstances...” (Thomas and Nelson 1996)*
 - *In other words, if you take the measure again, will you get the same result?*
- Valid
 - *“Degree to which a test or instrument measures what it purports to measure” (Thomas and Nelson 1996)*
 - *In other words, does your measure measure what is it supposed to measure?*

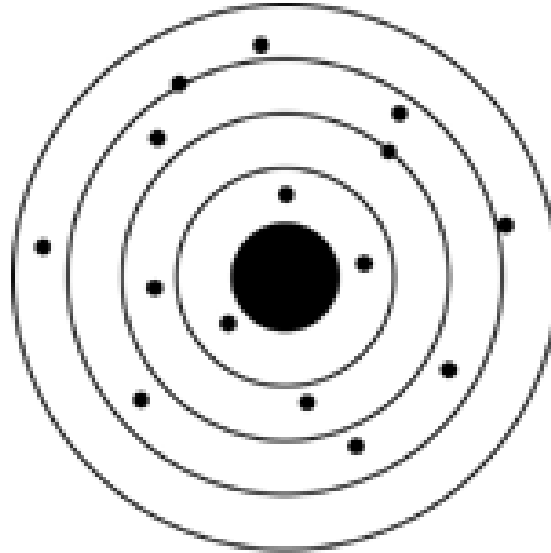
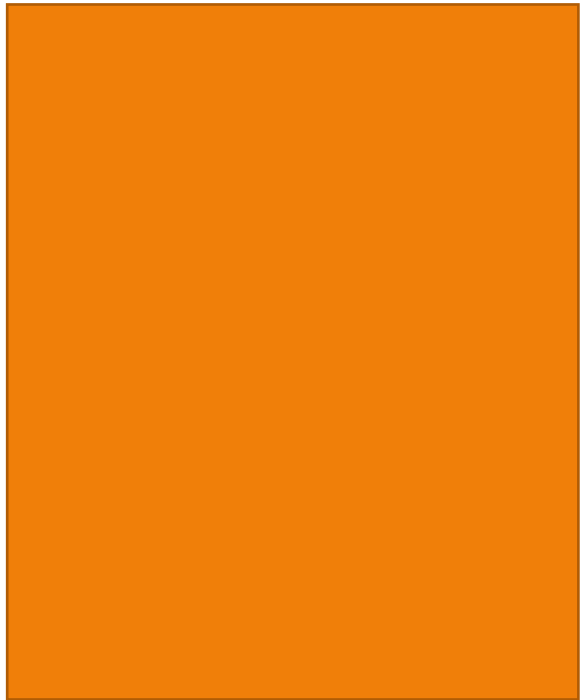
Illustration



Reliable but Not Valid



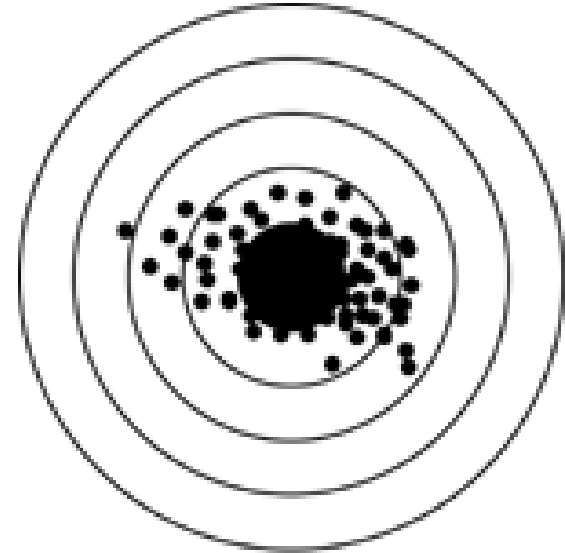
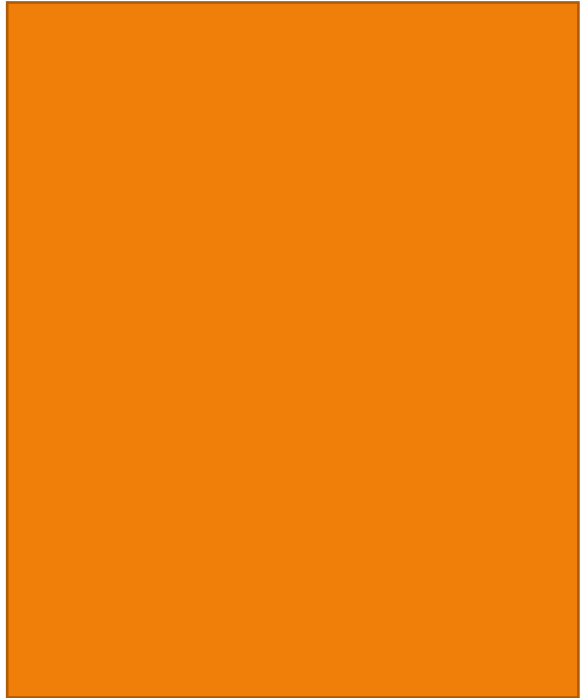
Illustration



Valid but Not Reliable



Illustration



Valid and Reliable

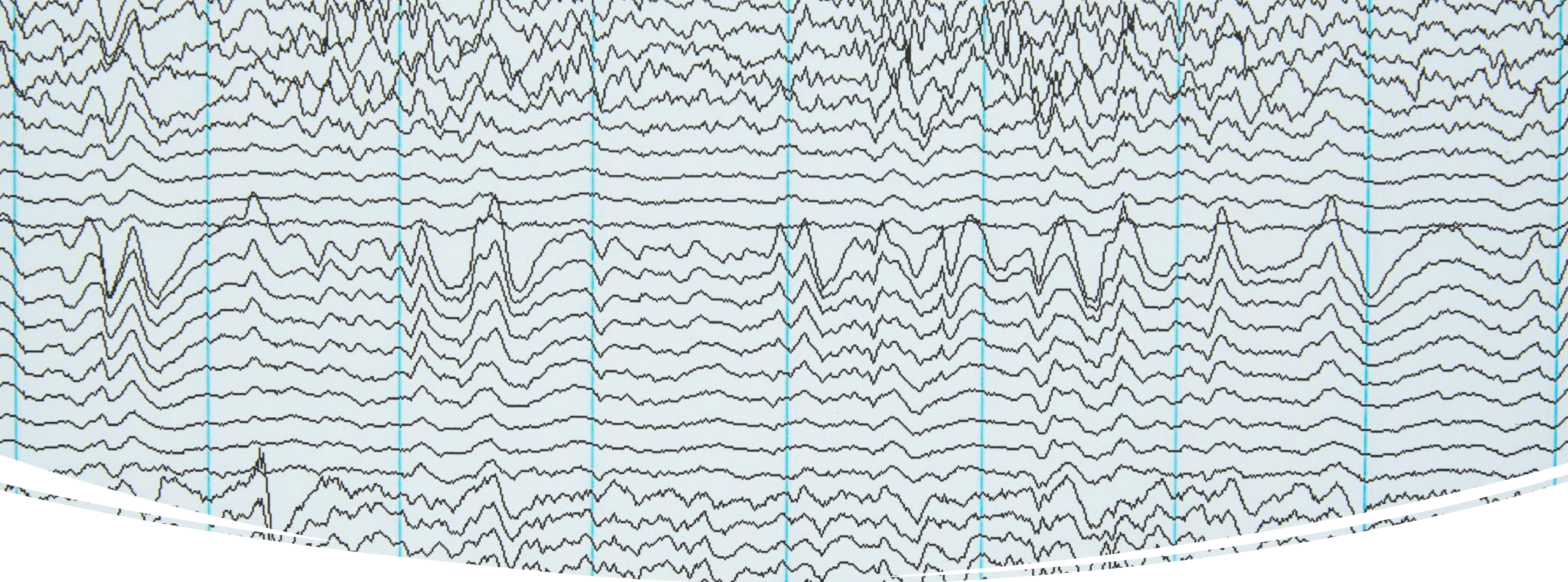


Summarize with Graphs



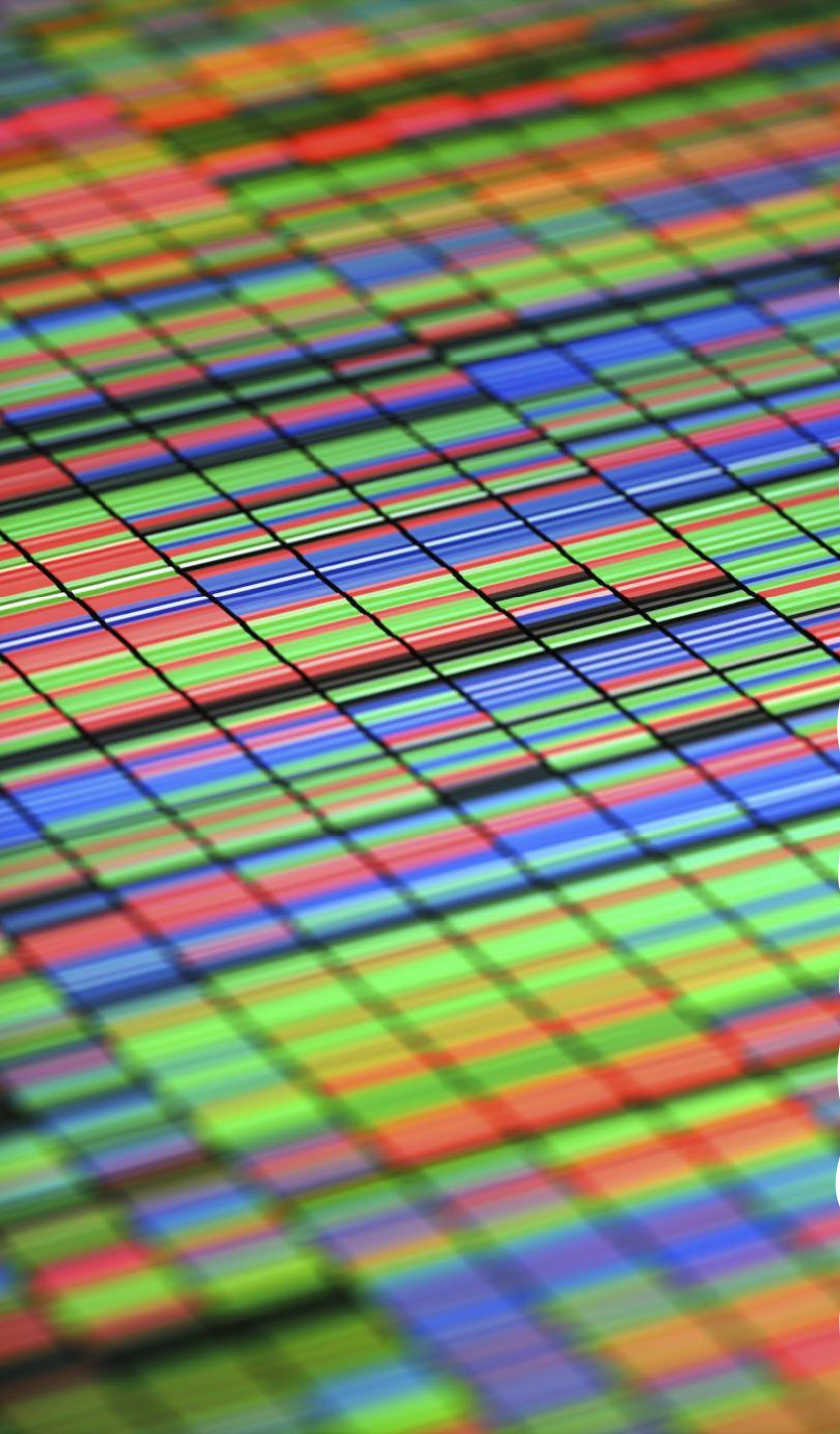
Summarize

- Transform a pile of numbers into a summary
- Descriptive Statistics
 - *Distribution of a variable is a table/graph showing the categories/values of outcomes and their frequency/percentage of occurrence*
 - *Exploratory Data Analysis (John W. Tukey 1977)*



Summarize

- Descriptive Statistics
 - Tables (example from: Woodbridge, 1845)
 - Graphs
 - Summary Statistics

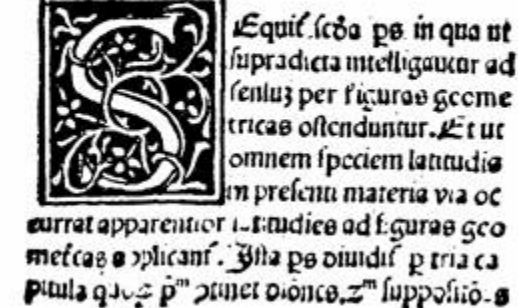


Categorical Variable Displays (Nominal, Ordinal)

- Frequency Distribution Graphs
 - Bar Chart
 - Pie Chart

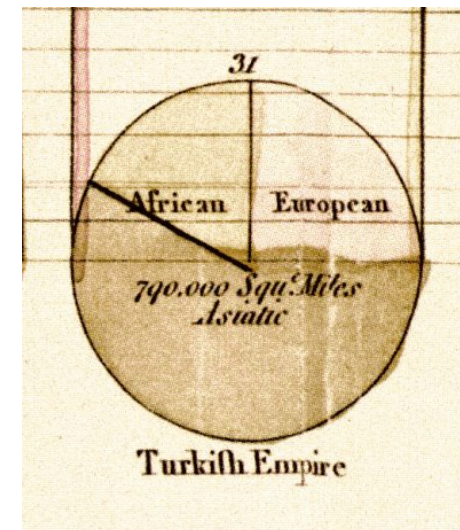
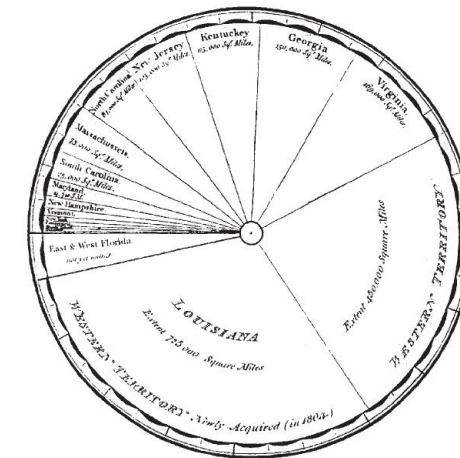
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A bar chart with 10 vertical bars of decreasing height from left to right. The bars are drawn with black outlines and are filled with a light gray color. The heights decrease steadily, with the first bar being the tallest and the last bar being the shortest.



Frequency distribution graph

- Pie Chart
 - Graphs of variables with categories of outcomes as frequency or percent of each category in the pie.
 - R Example
 - *A pie chart showing each state in the United States, part of Playfair's translation of A Statistical Account of the United States of America by D. F. Donnant.*
 - Earliest example seems to be from 1801

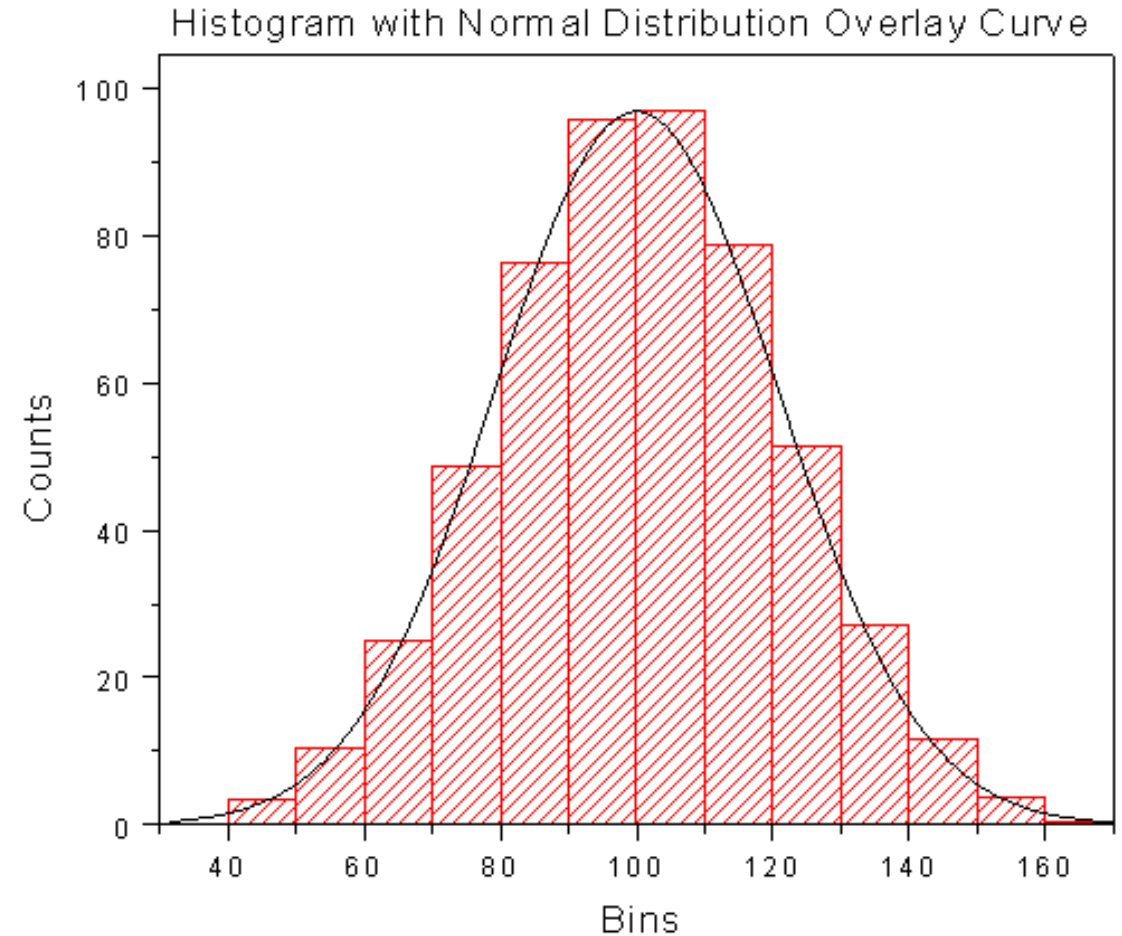


- Interval or Ratio Scales
 - *Histograms*
 - *Stem plots*
 - *Time plots*

Quantitative Variables

Histogram

- A histogram is a graphical representation of the distribution of numerical data.
- Approximates a probability distribution
- First described in Pearson, K. (1895).



Histogram

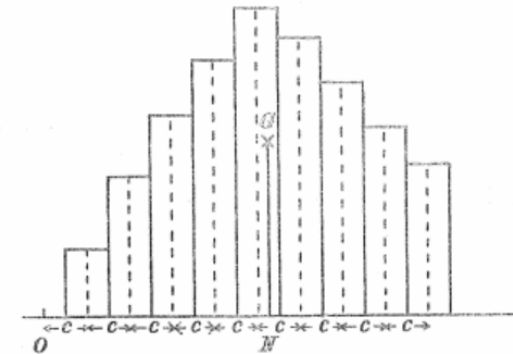
Pearson, K. (1895). Contributions to the mathematical theory of evolution. II. Skew variation in homogeneous material. *Philosophical Transactions of the Royal Society of London. A*, 186, 343-414.

boundary of the first rectangle, on the line of common bases, and let y_r be the height of the r^{th} rectangle, or

$$y_r = \frac{\alpha}{c} \frac{n(n-1) \dots (n-r+2)}{r-1} p^{n-r+1} q^{r-1},$$

while

$$y_1 = \alpha p^n / c.$$



Let us find the values of

$$\sum \{y_r c \times (rc)^s\},$$

15, 16, 21, 23, 23, 26, 26, 30, 32, 41

Stem	Leaf
1	5 6
2	1 3 3 6 6
3	0 2
4	1

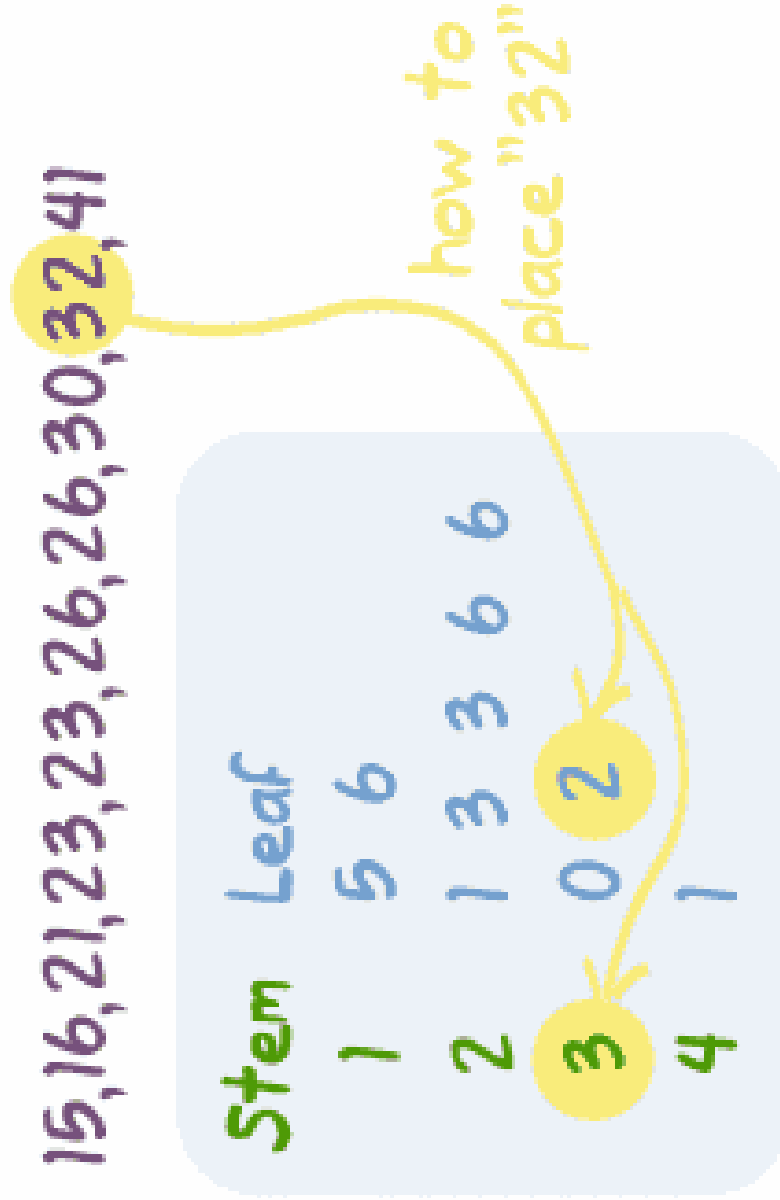
how to
place "32"

Stem plot

a.k.a. a stem and leaf diagram

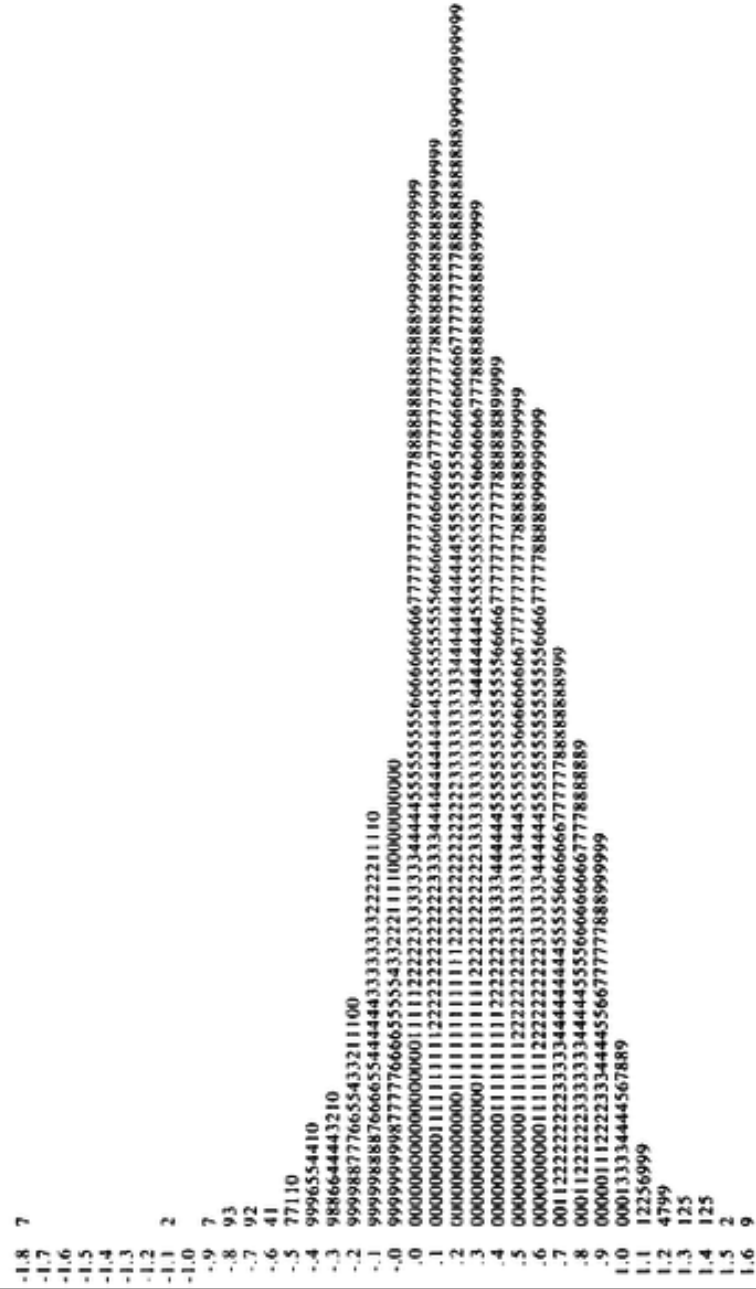
Stem plot

- a.k.a. a stem and leaf plot

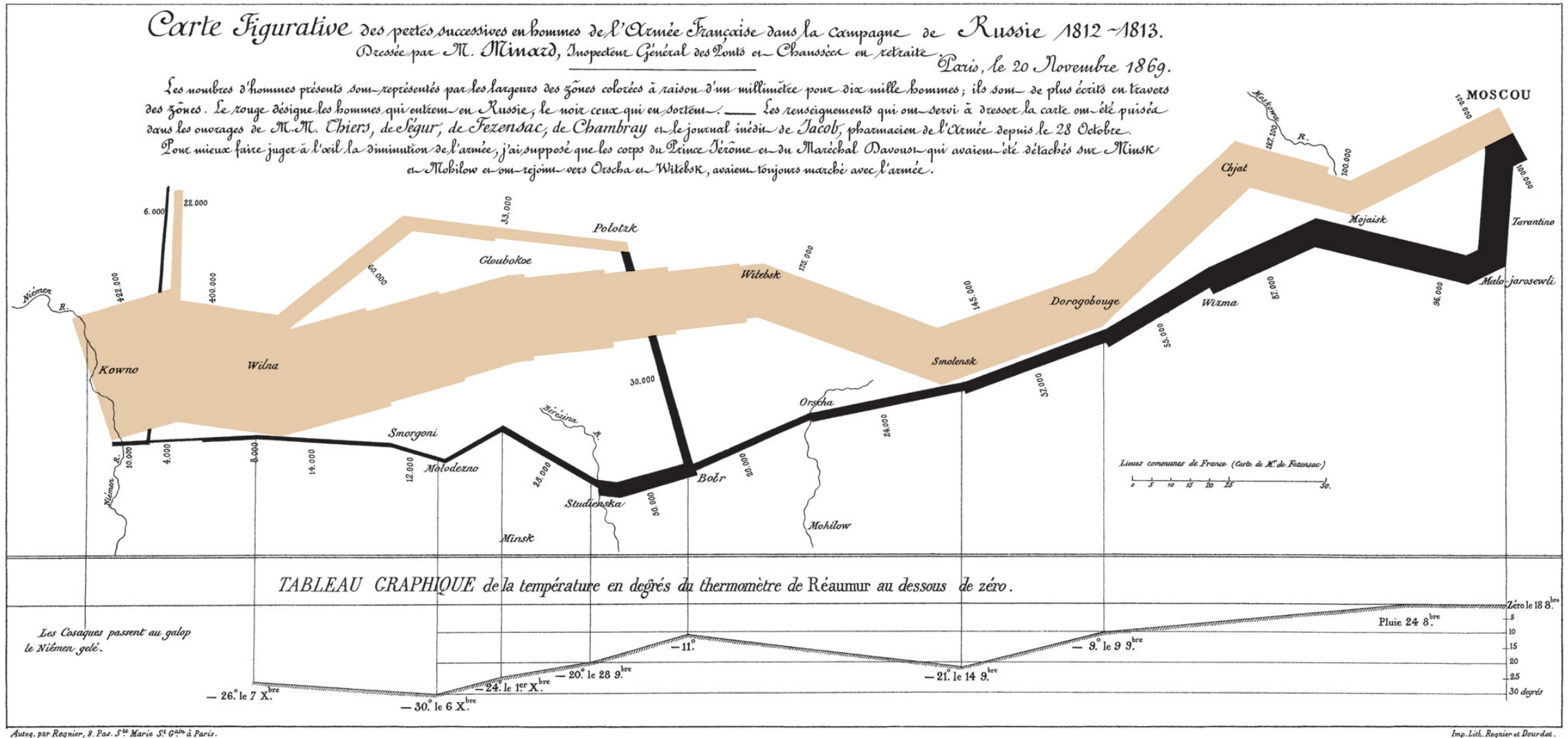


Stem plot

- a.k.a. a stem a



Time Plots



Napoleon's March

Information

Size of Napoleon's Grande Armée

Longitude of the army's position

Latitude of the army's position

Direction of the army's movement

Date of points along retreat path

Temperature during the army's retreat

Aesthetic

Width of path

x-axis

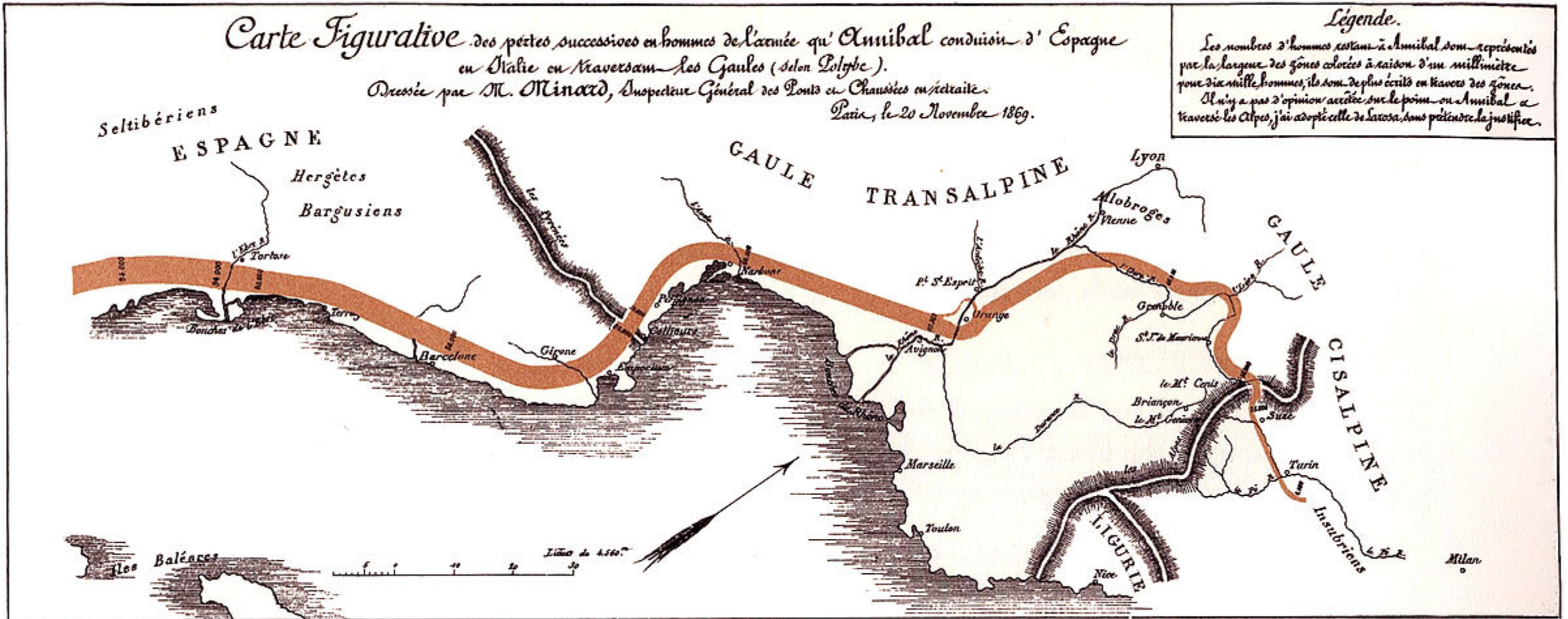
y-axis

Color of path

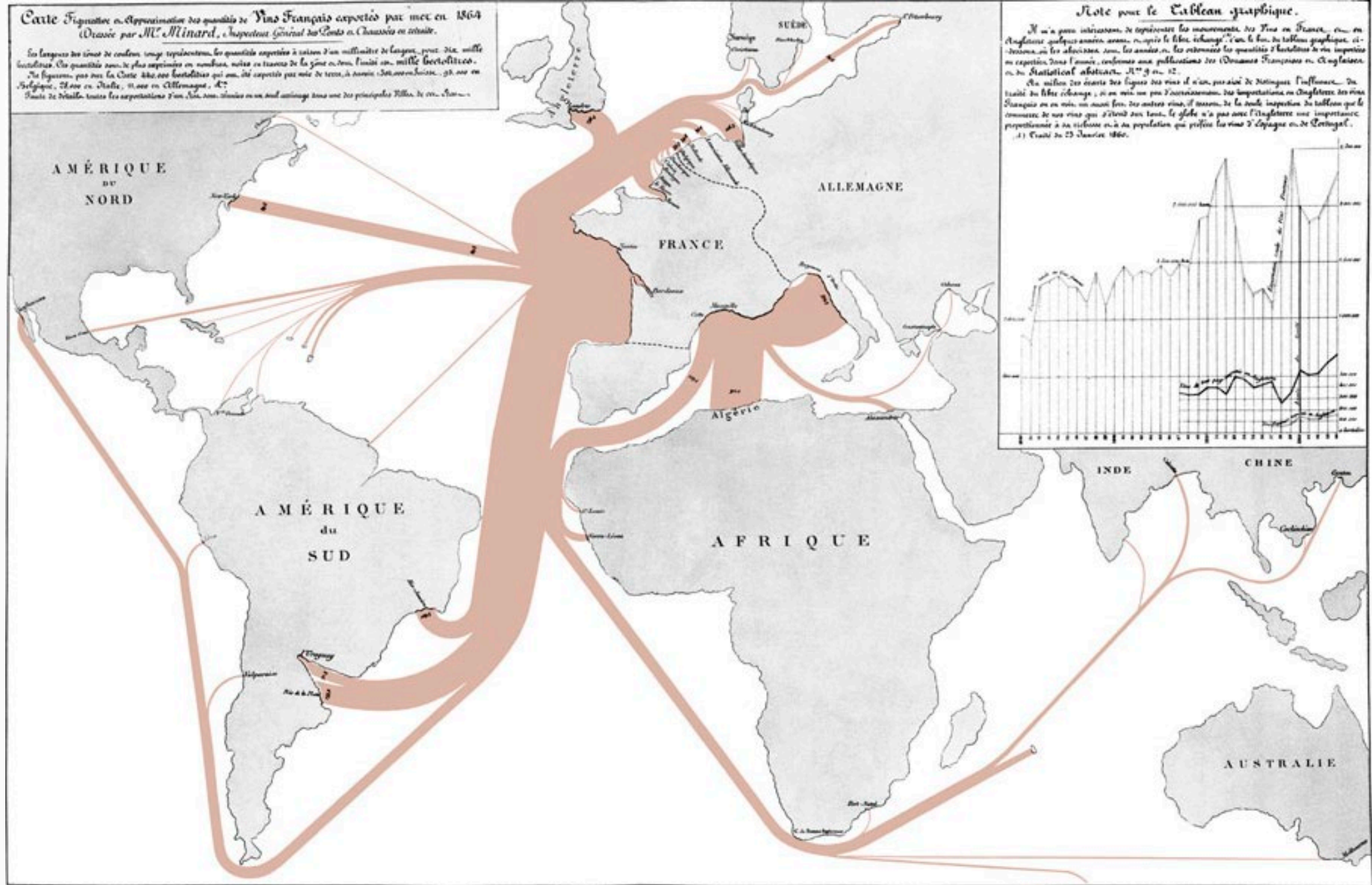
Text below plot

Line below plot

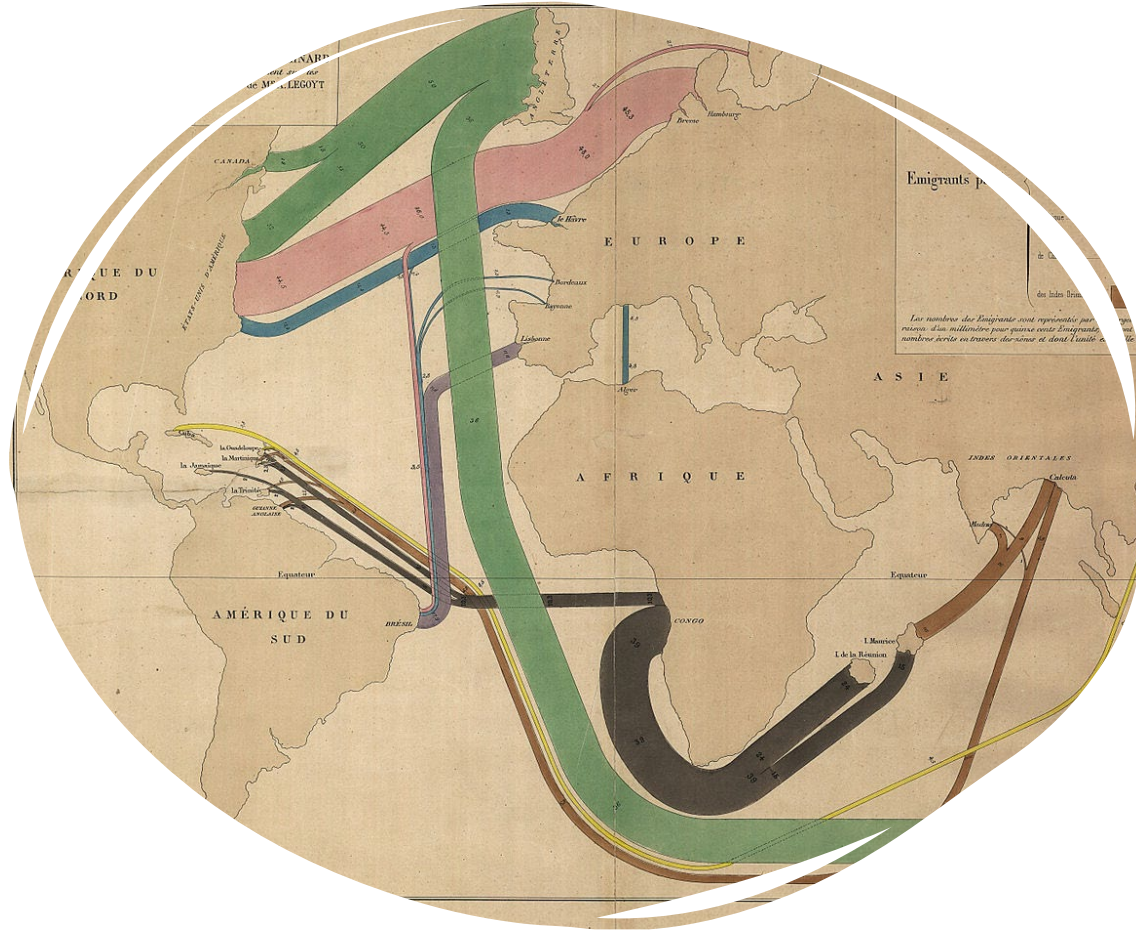
Other Data-Dense Graphs by Minard



- Men losses in the raid of the troops of Hannibal from Spain to Rome



- French wine exports for 1864



A map showing
worldwide
emigrants in
the year 1858.



Summarize with Numbers

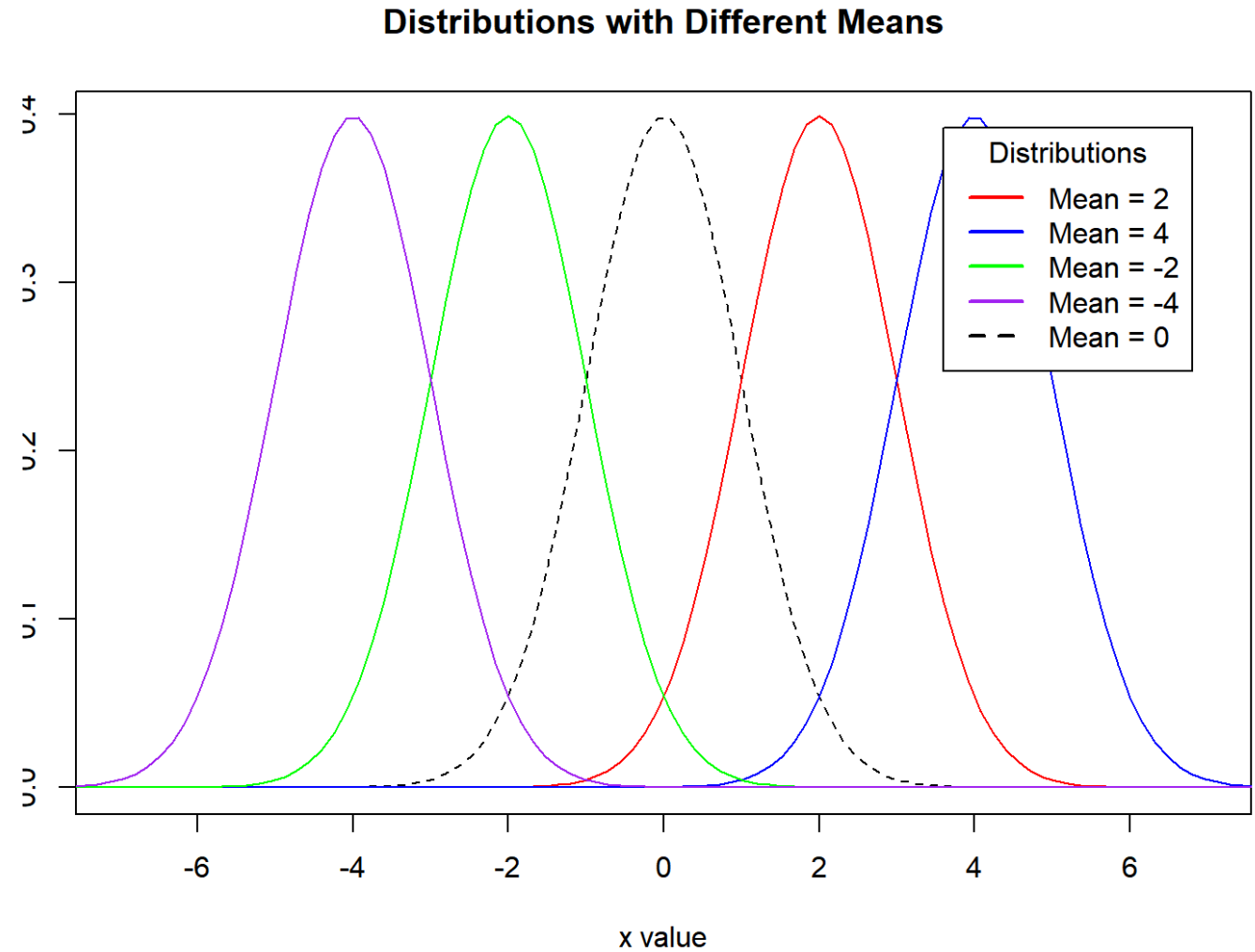


Characteristics of distributions

- Describing distributions with numbers
 - *Hundreds of descriptive statistics*
 - *Goal is to describe the data with a single number that represents an entire distribution*
 - *Taxonomy*
 - Central tendency / location
 - Spread
 - Skew
- 

Central tendency / location

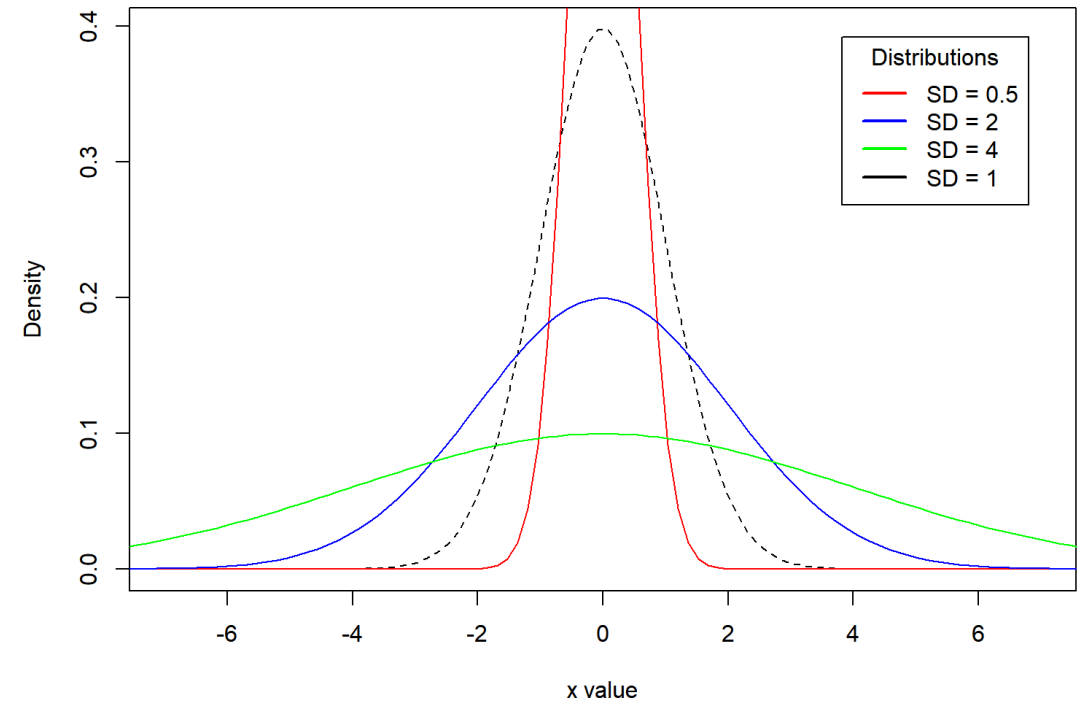
- Central tendency aims to capture the center of the distribution.



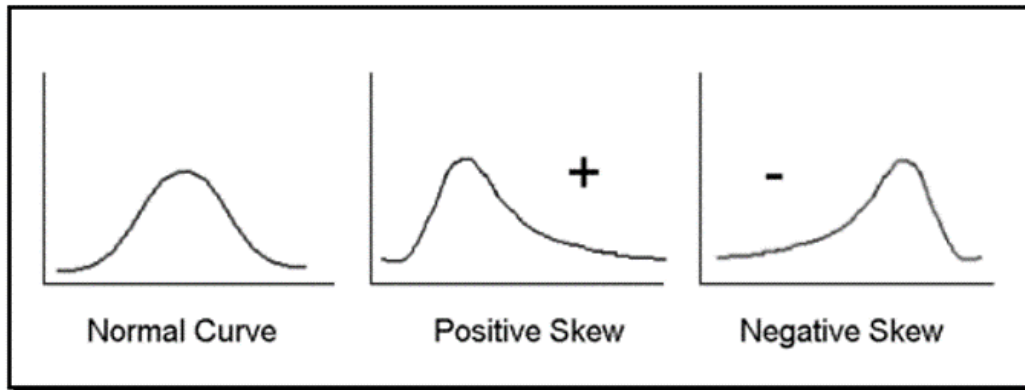
Variability/spread

- Variability describes how spread out the data are from that center
 - *Low variance has a less wide distribution, with the bulk of the mass in the center*
 - *High variance has a very wide distribution, bulk of distribution is spread out*

Distributions with Different Standard Deviations



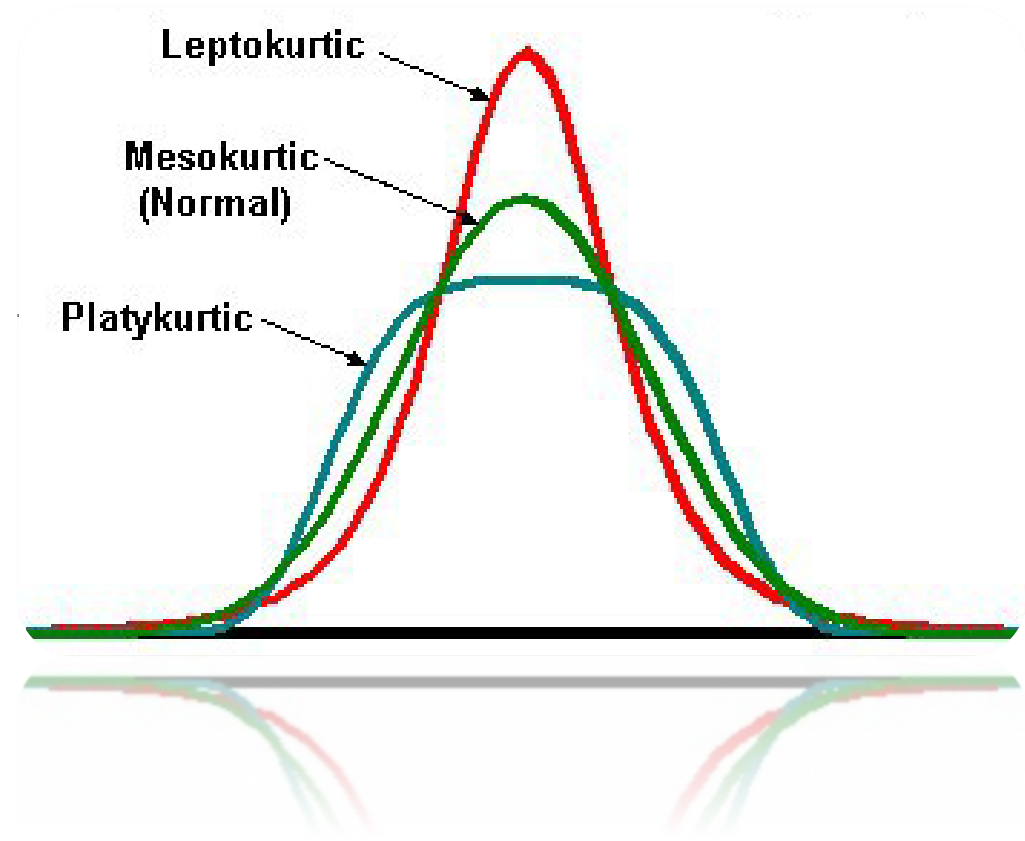
Skew/ Asymetry



- Skewness is a measure of symmetry, or more precisely, the lack of symmetry.
- A distribution, or data set, is symmetric
 - *if it looks the same to the left and right of the center point.*
- Refers to the tail of the distribution
- Positively skewed (right skewed)
 - *Bulk of the mass is on the left*
- Negatively skewed (left skewed)
 - *Bulk of the mass is on the right*

Kurtosis – peakedness

- Kurtosis is a measure of whether the data are
 - *heavy-tailed or light-tailed relative to a normal distribution.*
- Data sets with high kurtosis tend to have heavy tails, or outliers.
- Data sets with low kurtosis tend to have light tails, or lack of outliers.
 - *A uniform (flat) distribution would be the extreme case.*



Measures of Central Tendency

Mean

Median

Mode

Central Tendency: Mean

- Mean (μ ; $\bar{x}$)
 - $\bar{x} = \frac{1}{n} \cdot \sum_{i=1}^n x_i$
 - Properties
 - Is the balance point of the distribution (in terms of center of mass)
 - $\sum_{i=1}^n (x_i - \bar{x}) = 0$
 - Least squares property
 - The sum of squared deviations about the mean is small than for any other constant
 - $\sum_{i=1}^n (x_i - c)^2 = \text{minimum}$
 - highly sensitive to outliers (extreme scores)
 - (weakness; it means that the mean is not so good as a measure of central tendency in highly skewed distributions)
 - Is not a robust statistic (low robust = sensitive to outliers; high robust = not sensitive to outliers)

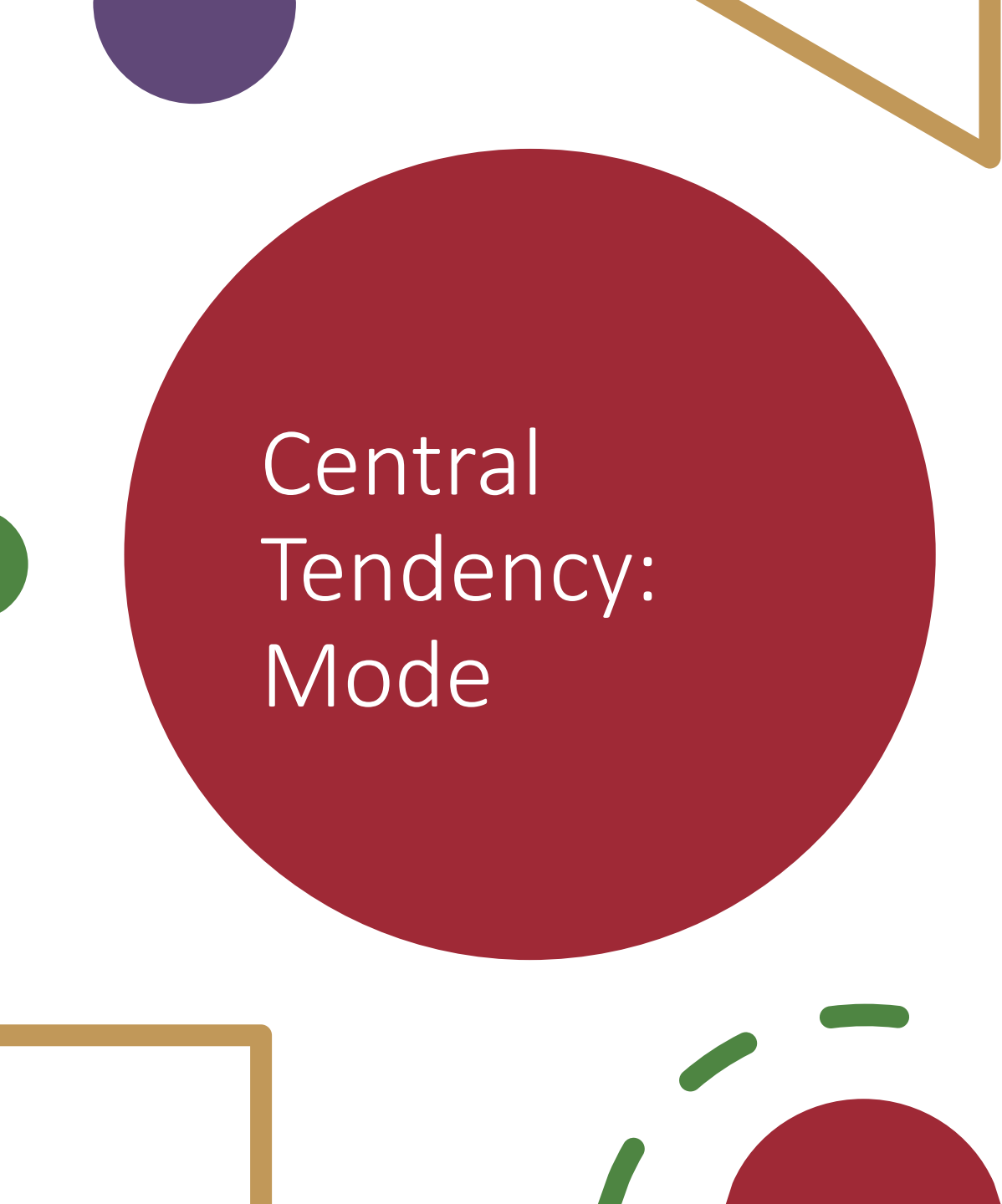
Mean

- Very good with quantitative data (interval and ratio data,
 - especially bell shaped distributions)
 - Very popular statistic



Central Tendency: Median

- Median (Md)
 - If **n is odd** then
 - Med = value of the $\frac{n+1}{2}$ item term.
 - If **n is even** then
 - Med = average of the $\frac{n}{2}$ and $\frac{n+1}{2}$ item terms.
 - Highly robust to outliers (less sensitive than the mean to outliers)
 - Properties
 - Balance point of scores
 - Sum of absolute deviations is smaller than any other constant
 - $\sum_{i=1}^n |x_i - c|$ is minimized
 - Often used for ordinal data

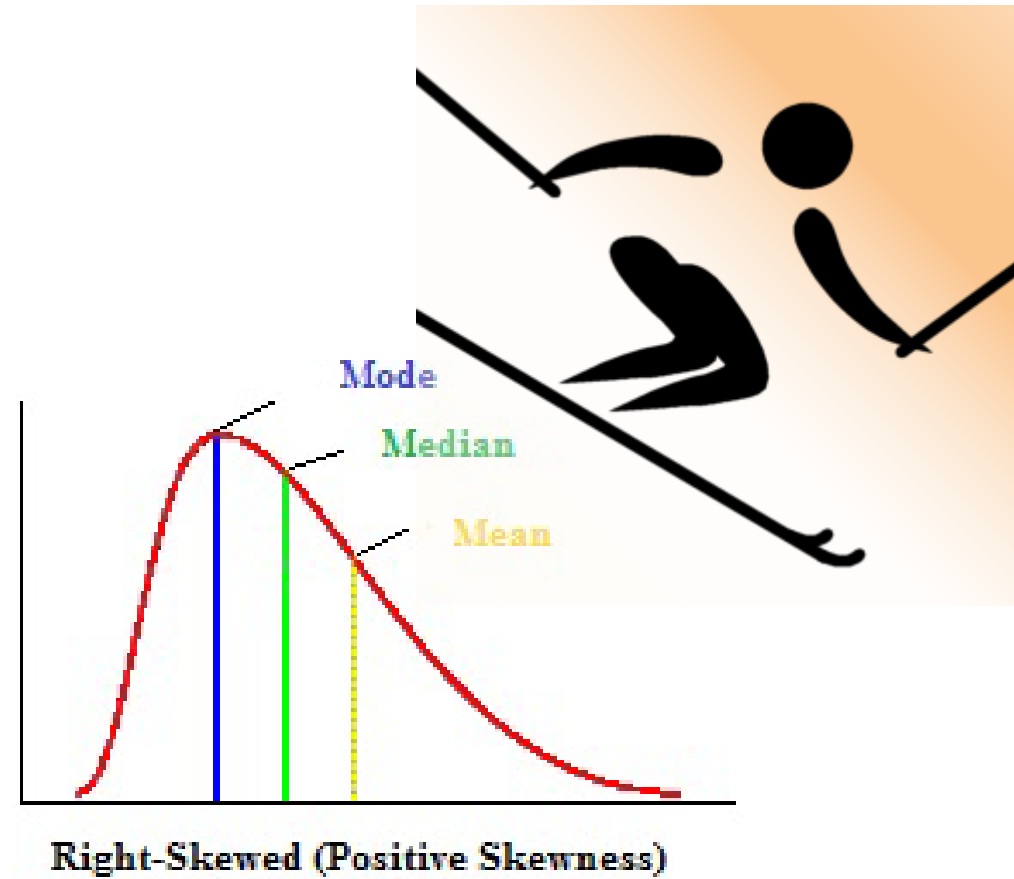
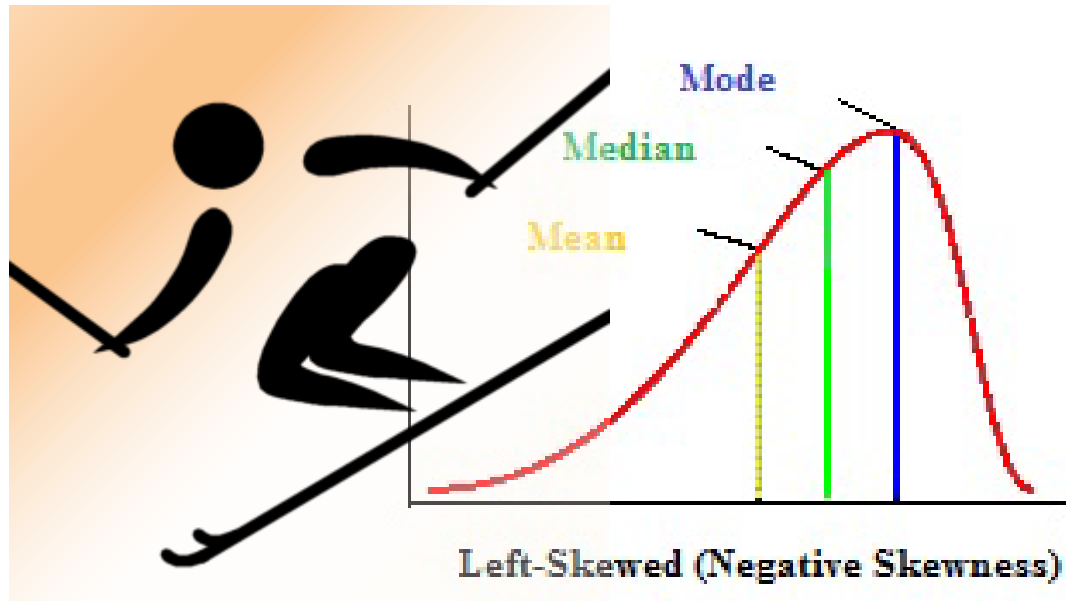


Central Tendency: Mode

- Mode (Mo)
 - The most frequent value
 - Local modes are the highest point with a subset of the distribution,
 - there can be multiple ones

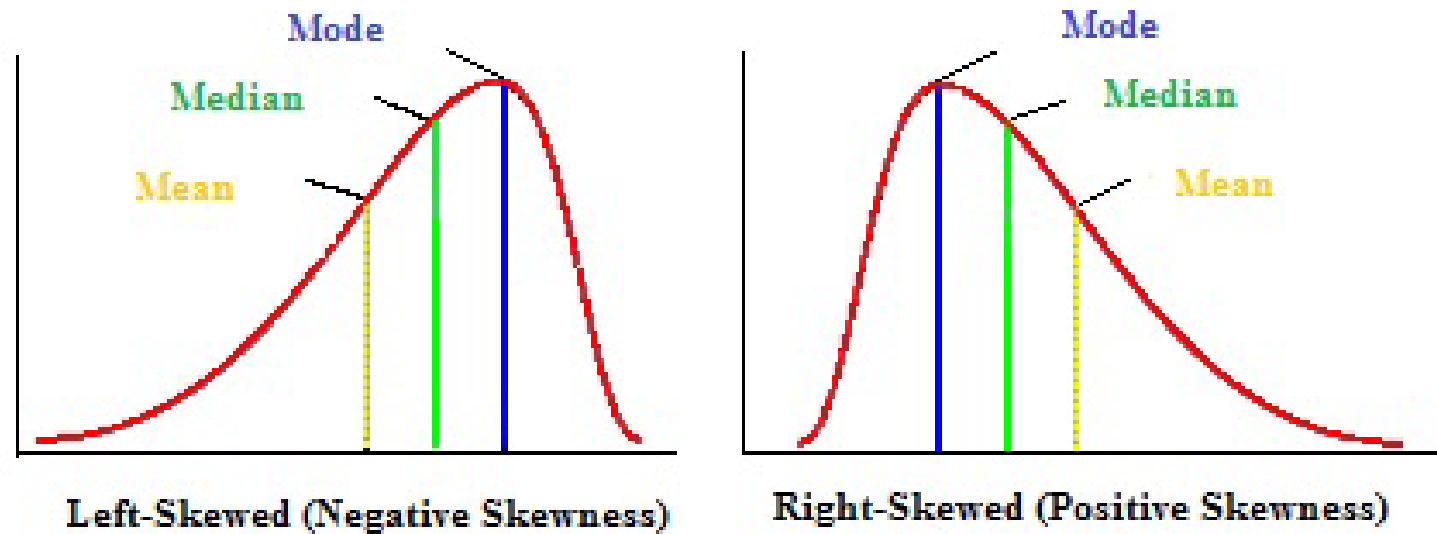
Skewness

- Cowardly Skier



Relationship between mean, median, and mode

- Symmetric unimodal distribution
 - Mean=median=mode



Positive Skew

The distribution is said to be *right-skewed*, *right-tailed*, or *skewed to the right*,

- *despite* the fact that the curve itself appears to be skewed or leaning to the left;

The right tail is longer;

the mass of the distribution is concentrated on the left

right instead refers to the right tail being drawn out and, often,

A right-skewed distribution usually appears as a *left-leaning* curve.

Negative Skew Facts

Also called: *left-skewed*, *left-tailed*, or *skewed to the left*

The left tail is longer;

the mass of the distribution is concentrated on the right of the figure.

left instead refers to the left tail being drawn out

A left-skewed distribution usually appears as a *right-leaning* curve.

Spread around the Median

- Range
 - $Max(x_i) - Min(x_i)$
 - Non-robust to outliers
- Quartiles
 - Lower quartile (Q1), 25th percentile
 - Second quartile (Q2), 50th percentile / median
 - Upper quartile (Q3), 75th percentile
- Interquartile Range (IQR)
 - Q3-Q1
 - Sometimes called h-spread; h = hinges

Spread around the Mean

- Variance (σ^2 ; s^2)
 - Measure of spread around the mean
 - Goal of the measure to use every score
 - $\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$
 - $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$
- Standard Deviation (σ ; s)
 - $\sigma = \sqrt{\sigma^2}$
 - $s = \sqrt{s^2}$
- s^2 is nearly the average squared deviation
- s^2 uses n-1 instead of N
 - This adjustment is called Bessel's correction
- Properties
 - Std is in raw score units
 - Var + std dev are highly NOT robust

Normal Distributions and Rescaling

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Normal Distribution

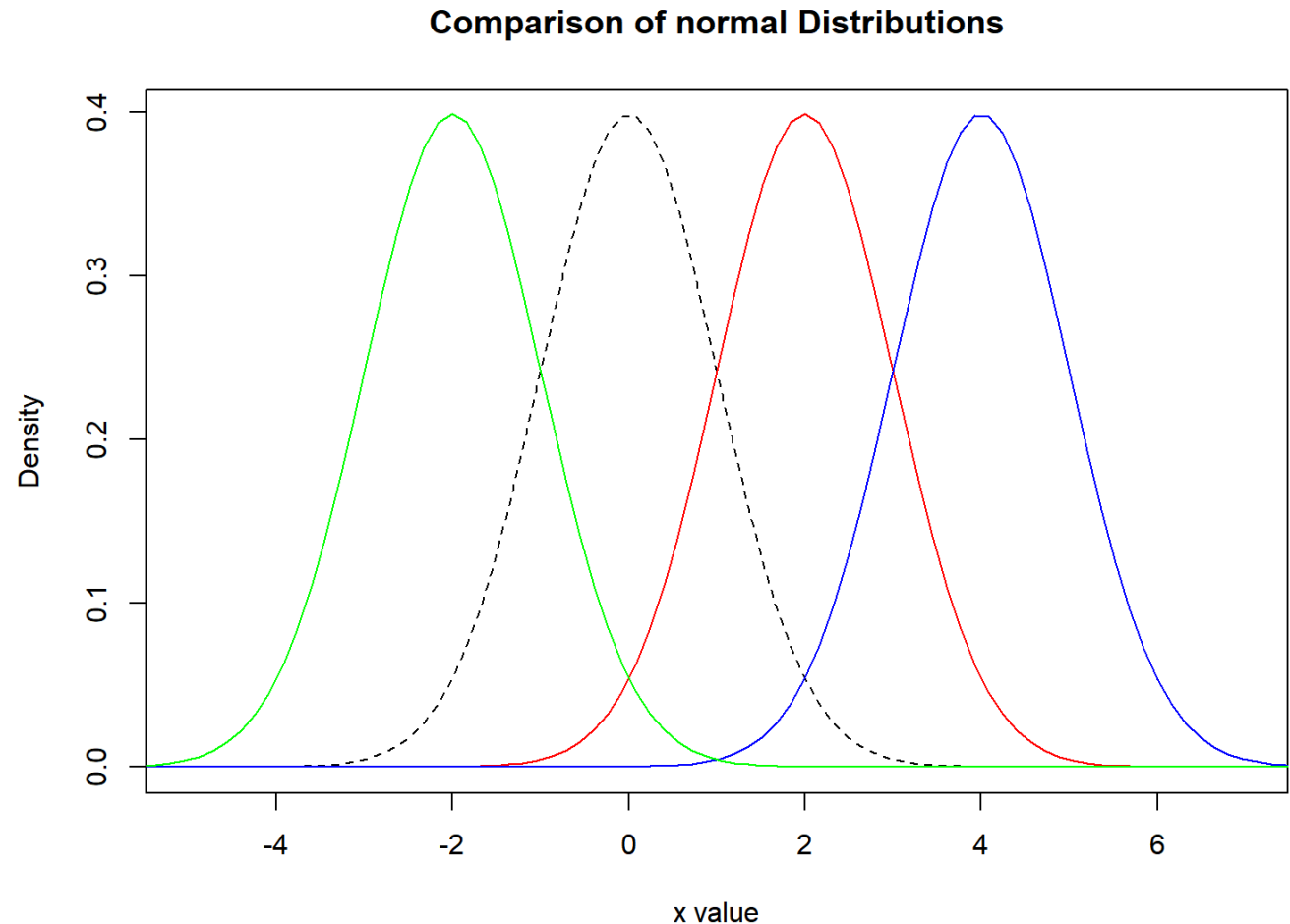


$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

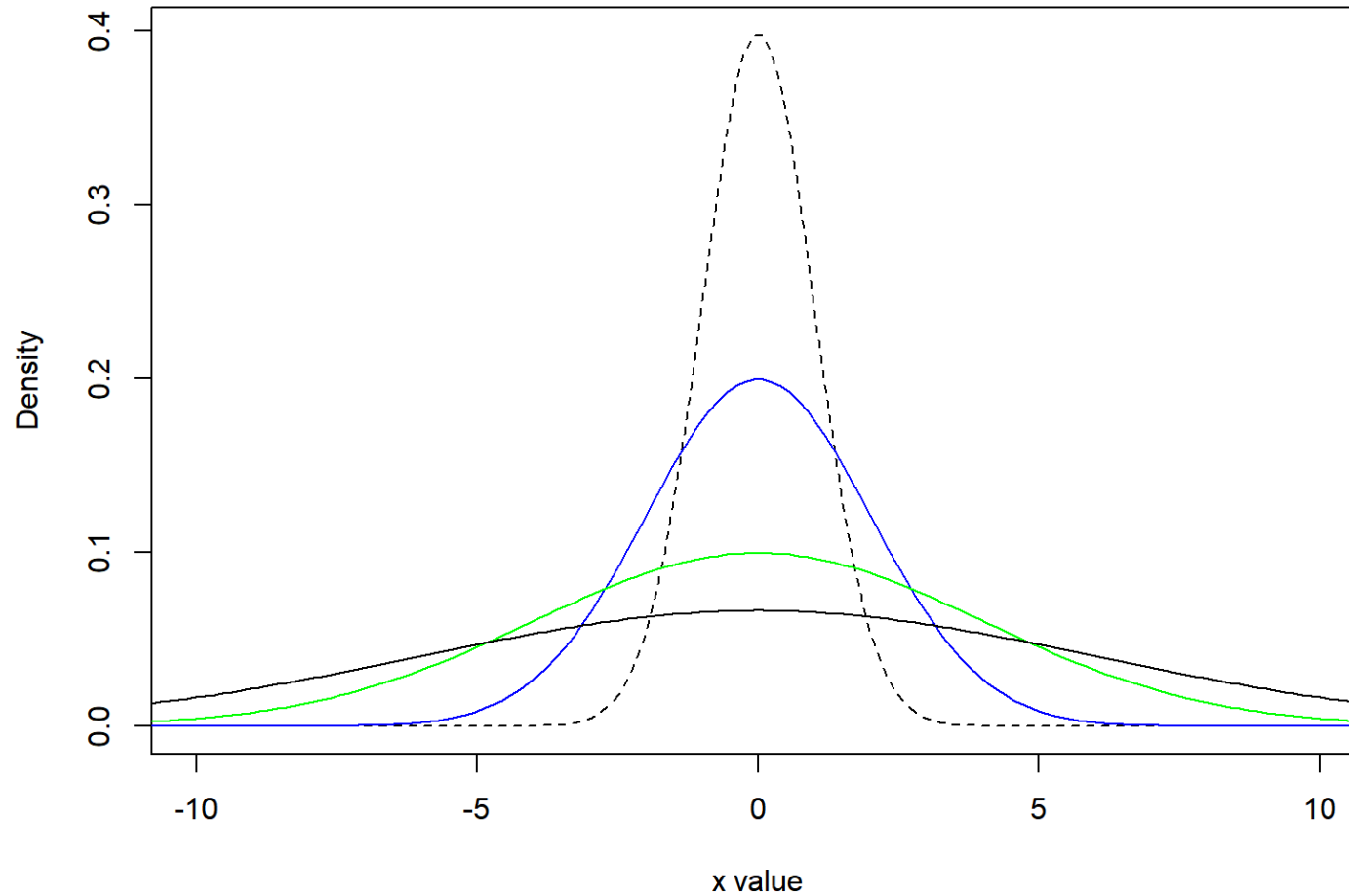
- Def – a particular bell-shaped curve that has the following mathematical properties
- Formula has two parameters
 - mu
 - sigma
- The standard normal (mu=0; sigma=1) simplifies the equation

Plots of the Standard Normal with multiple means

- The mean is located at the center of the symmetric curve and is the same as the median.
- Changing μ without changing σ moves the Normal curve along the horizontal axis without changing its variability.



Comparison of normal Distributions

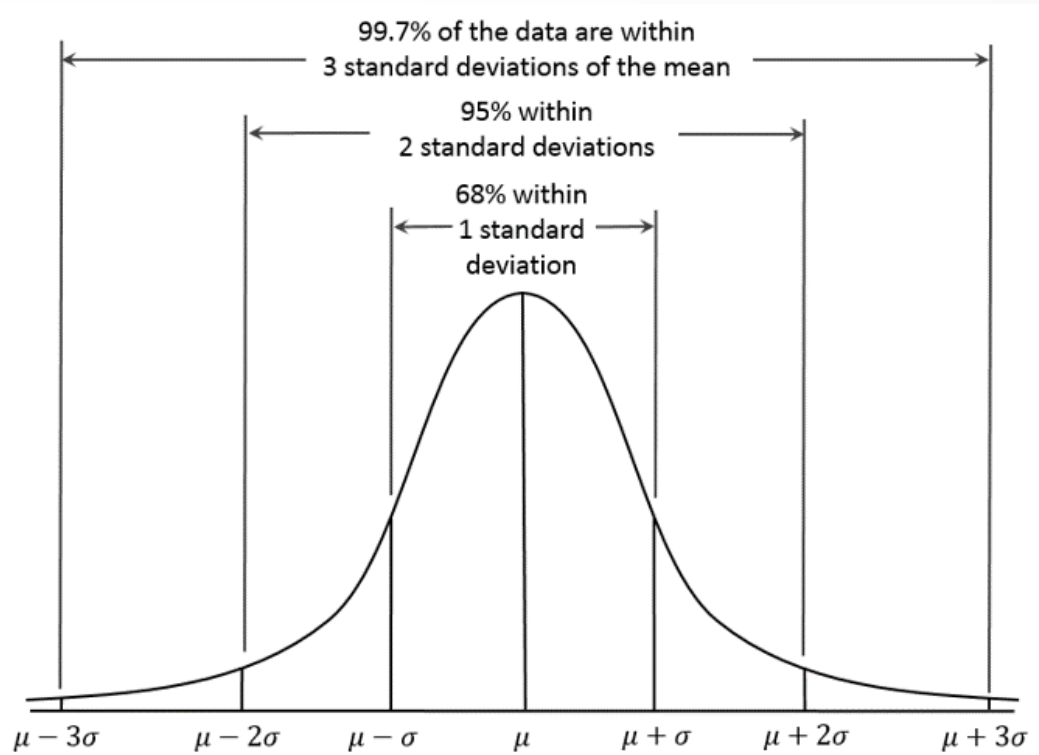


Plots of the Standard Normal with multiple standard deviations

- The standard deviation σ controls the variability of a Normal curve. When the standard deviation is larger, the area under the normal curve is less concentrated about the mean.
- The standard deviation is the distance from the center to the change-of-curvature points on either side.

The 68-95-99.7 Rule

- approximately 68% of the observations fall within 1σ of μ
- approximately 95% of the observations fall within 2σ of μ
- approximately 99.7% of the observations fall within 3σ of μ



Normal distribution tricks

Symmetric

50% of area above zero

Total proportion is 1.0 (or 100%)

Rescaling

All Normal distributions are the same if we measure in units of size σ from the mean μ as center.

We can convert any variable into the same metric as the standard normal

Changing to these units is called standardizing or rescaling.

Rescaling

- Converting Statistics Sample
 - $Z_i = \frac{x_i - \bar{x}}{s}$
- Population
 - $Z_i = \frac{x_i - \mu}{\sigma}$
- Notes: Skewed distribution -
> transformed into z score
 - Keeps the exact same shape.
 - Only the numbers change (mean, sd etc).
- Advantages
 - Allows us to compare scores on a common metric
- Origin is 0. The mean
 - The units are 1, the standard deviation
 - +-> above the mean
 - - => below the mean
- We can compare across measurement scales
- Shape of the distribution does NOT CHANGE
- We can go from z-scores to raw scores

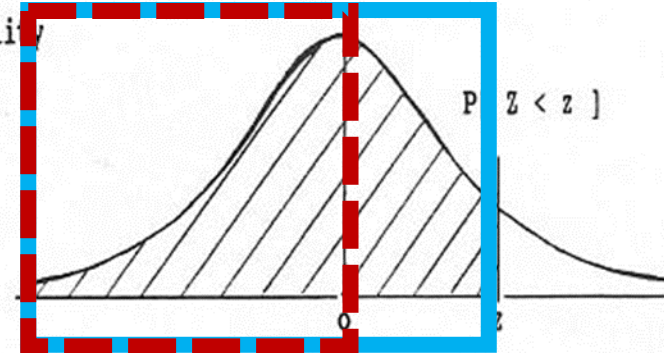
Worked Z-Score Problem

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability
up to the standardised normal value z
i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2) dz$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	.8413	.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319



Comments on Standardization

- Standardization of variables
 - Transformation – ratio variable
 - Features of measurement scales
 - Some have built in properties
 - Percentiles (not equal interval!)
 - Standardized scale (z score scale)



Percentiles

- Percentiles (percentile rank)
- Score value that cuts off a certain % of scores below it (cumulative percentile)
- Special percentile ranks
 - M: 50%
 - Q1: 25%
 - Q3: 75%
- Don't have equal intervals
 - Weak statistically



\$45,000



\$15,000



\$10,000



← **ARITHMETICAL AVERAGE**

\$5,700



\$5,000



\$3,700



← **MEDIAN** (the one in the middle)
(12 above him, 12 below)

\$3,000



\$2,000

← **MODE**
(occurs most frequently)

Robustness

- Robust Statistics are less sensitive to outliers
- Most statistics, like
 - means, standard deviations, and correlations, and
 - every statistic based on these, are highly sensitive to outliers.